Analog Design of an AI-Powered Synthesizer

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by

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Abstract

This writing discusses the background, design, testing, and implementation of a fully analog audio signal processing chain. This chain consists of a 4-band equalizer, phaser, tremolo, clipper, and wah-wah, the order of which is permuatble by patch cabling. This chain is meant to serve as the post-processing for the *Waveform.ai* project AI-powered synthesizer. Engineering and appropriate design for low-frequency oscillators and voltage-controlled resistors are discussed, as well as design of the above-mentioned processing effects. Lastly, the system-level integration of all circuits is explored. Analog signal processing was found to demonstrate exceptional capability accompanied by unique sonic characteristics which distiguish it from digital signal processing. Although many audio effects may be implemented practically in the analog domain, some others—especially time-based ones—are far more difficult and require specialized fabrication.

Introduction

Waveform.ai is a multidisciplinary project supported by the William & Mary Arts and Science Exchange (ASE). The project consists of an ambient, AI-driven synthesizer improvisation, a sensor-augmented flute performance, and concludes with a interactive live-coding finale. The artistic themes of the project revolve around the fluency of water and the machine as an artist. This writing discusses the analog signal processing design for the synthesizer used in the first movement of the performance. Synthesizers are a category of musical instruments which generate sound through electronic audio signals. Typical synthesizers offer the musician a table of wave shapes—which dictate the initial timbre of the sound—and further control over sonic characteristics through analog or digital wave-shaping effects. The *Waveform.ai* synthesizer advances user control by implementing artificial intelligence in the process of sound design. By providing the synthesizer with a language-based description of the a sound—for example "warm, soft, and short"—they will receive a tone which best matches the described sonority. Such functionality relieves the user from the steep learning curve associated with sound design, opening the world of synthesis to beginner musicians. After a desriable tone is found, it is produced digitally by a Raspberry Pi running the Dexed Yamaha DX7 digital synthesizer. This digital waveform is converted to an analog signal, which the user may further modify it with a permutable chain of analog effects. Before digital signal processing became ubiquitous, audio signal processing was performed using analog circuitry. Analog circuitry brings with it many limitations and quirks, and these quirks become beloved by the audio engineering community—even long after digital signal processing became standard. Analog processing is often described as colorful and rich, imparting sonorities to audio that are sorely missed in digital processing. For this reason, the majority of post-synthesis processing of the signal chain was chosen to be performed by analog circuitry.

This signal chain was initially chosen to consist of an equalizer, phaser, wah-wah, reverb, delay, clipper, tremolo and chorus. These effects were chosen for their user friendliness

(easily audible to the untrained ear) and thematic connections to the artistic aesthetic of *Waveform.ai*. Many of them are characterized by slow, undulating modifications to the spectral content and envelope of a sound, while others allow for the transformation of a natural sound to one reminiscent of industry and machinery.

Project Objectives

The analog design of the synthesizer must achieve the following mission:

- The ease of use for a beginner should be prioritized
- There should be modularity in the analog signal chain, meaning a user may select any order of effects to compose a signal chain
- The analog signal processing should allow for vast control over a sound
- A user should not be confused about the audible effects of any of the processing—in other words, they should be able to hear how they are changing the sound.

Circuit Design

Analog circuits consist of electrical connections between components such as resistors, capacitors, inductors, diodes, and transistors. Simple circuits may be directly soldered onto prototyping boards, however as the wiring complexity and number of components in a circuit increases, this becomes exceedingly difficult. Instead, components are soldered onto printed circuit boards (PCBs) which contain through-hole connections connected by machine-drilled copper traces. By directly printing wiring connections, the physical complexity of the circuit board is greatly reduced.

The electronic computer-aided design software Kicad was used to design these PCBs. Kicad allows for circuit schematic capture, generation of gerber files, and 3D model viewing for designed PCBs. The software uses components' physical footprints (ie. resistor sizes, IC packages) and a schematic of a circuit to generate wire traces on a PCB. Custom footprints may be drafted in the software for unconventional components.

Multisim is a SPICE-based (Simulation Program with Integrated Circuit Emphasis) electronics simulation software, which allows for time and frequency domain graphing of simulated circuits. This tool was an essential component of the circuit design process, as it allows for easy probing and demonstrates idealized circuit component behavior. Although Multisim offers vast capabilities, including simulations of common named components (such as the ua741 op-amp and 1N4148 diode), many components are missing, often requiring breadboard prototyping in place of software simulation.

Low-Frequency Oscillator

Many audio processing effects rely on the periodic, time-based modulation of specific elements of their transfer functions— such as gain, phase, or cut-off frequency. Therefore, it is necessary to build a circuit which may electrically control a parameter of the circuit in a periodic manner; this is the purpose of a low-frequency oscillator (LFO). For the circuits in this project, a relaxation oscillator topology was used because of its simplicity, tunability, stability, and triangular output wave shape.

- Simplicity: the relaxation oscillator is built with fewer than 10 components, all of which are standard in any electronics lab (op-amps, resistors, capacitors).
- Tunability: the oscillator's frequency is adjusted by a single potentiometer, meaning any desired frequency within a set frequency range may be achieved.
- Stability: oscillation is stable at any frequency, and requires only op-amp feedback to stabilize (compared to the Wein bridge oscillator, which often incorporates a light bulb load to apply a dynamic feedback resistance). During initial testing, a Wein bridge oscillator was built, however due to its complexity and relative instability, it was not implemented in any audio processing circuit.
- Wave shape: the LFO shape is an exponential RC charge-discharge oscillation, and depending on the particular biasing of the oscillator's switching threshold, the exponential wave shape may become triangular. This makes for a more interesting auditory effect— as opposed to a square or sinusoidal wave shape, which are respectively too jarring or too regular[1].

The relaxation oscillator is an elegant oscillator design based upon the Schmitt Trigger comparator. An RC network is then connected across the output of the op-amp to the

[1]A square LFO shape would essentially create an "on and off" effect for whatever parameter it is modulating, without any smooth transition between the two points. This could be interesting, however its use would be very niche. A sinusoidal LFO shape suffers from the opposite problem: it has very little "bite", and gives an excessively smooth traversal over a parameter

inverting terminal and then ground. A small instability in the circuit (noise, interference, floating pin voltages, etc.) will cause the comparator to flip to HIGH output, causing the capacitor to charge, eventually setting the inverting terminal voltage past the threshold, flipping the comparator output to LOW, and the oscillation continues. The output voltage of this oscillator was chosen as the voltage across the capacitor at the inverting op-amp terminal. This high-impedance output is then buffered and used to drive an external circuit.

Because this oscillator is used in all the following circuits to drive LEDs, its base topology was modified to allow it to swing between 2-4V. This ensures that the dynamic range of the control signal directly corresponds to the dynamic range of the LED brightness (without a large portion of the period sitting beneath the LED on voltage). This was done by applying a DC bias to the comparator crossing threshold, and limiting the rail voltages of the op-amp to 0V and 5V (with a 5V regulator (78M05BT) to step down the 15V supply). While such a design works in theory and in simulations, it failed in practice when built on a ua741 op-amp. This is due to the difference between supply voltage and rail voltage on the ua741, which sets a lower bound on op-amp output voltage to 2V rather than 0V. This prevents the output from triggering the comparator, rendering oscillation impossible. To remedy this, a rail-to-rail op amp is necessary, and the single-supply capabilities of the MCP6002 made it an ideal candidate. Additionally, the second op-amp in the package is used to buffer the output voltage and isolate the oscillator from whatever it may be used to drive. The individual resistance and capacitance values must be adjusted to fit the needs of the circuit the LFO is driving.

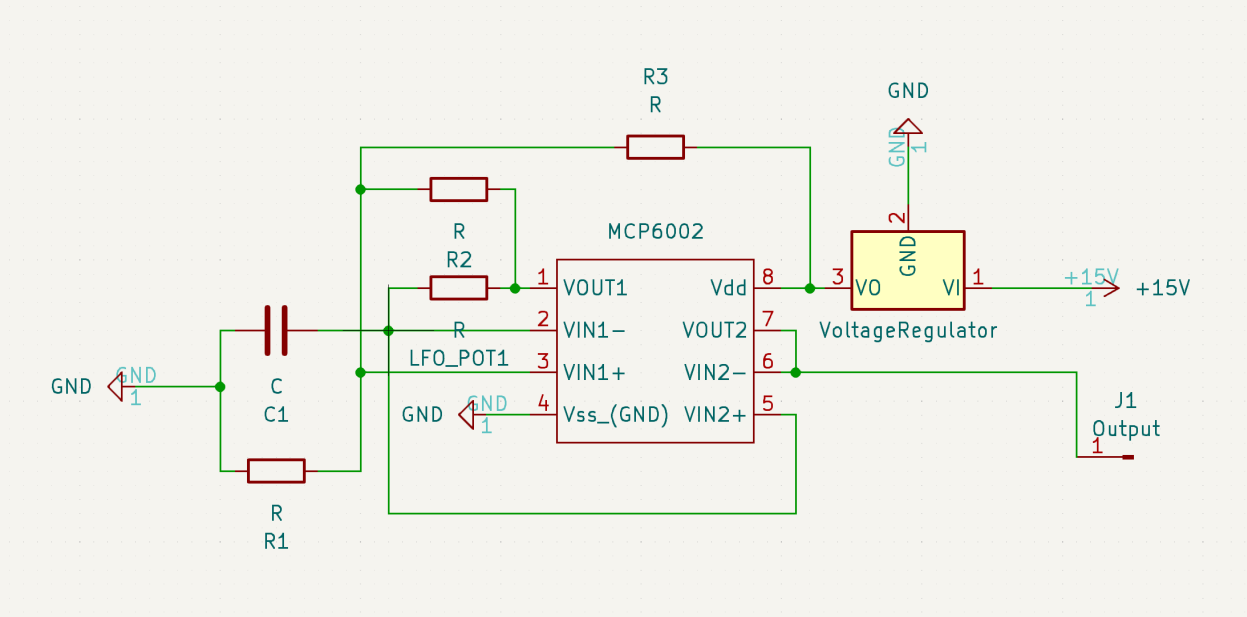

Figure 1: Circuit diagram for relaxation-based LFO

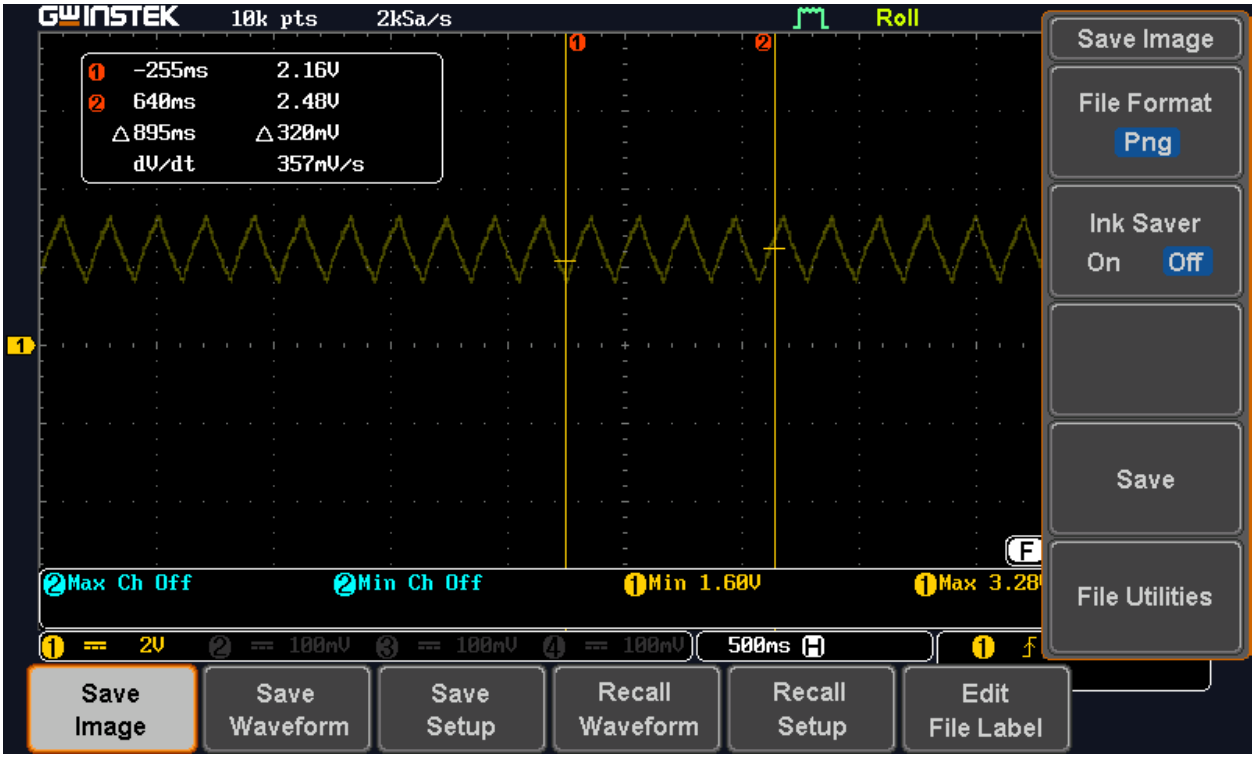

Figure 2: Example LFO output waveform

Voltage-Controlled Resistors

Once an LFO is built, using its control signal to modulate a circuit parameter is not trivial. Since the LFO produces a voltage, it requires a component whose behavior can be influenced by that voltage. Most passive components—such as resistors, capacitors, diodes, and inductors—have fixed properties determined by their physical design. Yet, important circuit characteristics such as gain and filtering directly depend on these components. Therefore, it is necessary to find a device that can act as a voltage-controlled equivalent to one of these standard components.

The first choice for such a device is the field-effect transistor (FET). A transistor is a three-terminal semiconductor device which controls the flow of current across two terminals based on a signal at a third terminal. The FET is a type of transistor which controls current flow through an electric field generated by a very small capacitor at its third terminal, known as the gate. This means the FET is controlled by a voltage at a high-impedance input. Under general operating conditions, the FET is non-linear component, meaning it does not obey Ohm's Law. However, it is possible to bias a FET such that it operates in a region where an increase in voltage results in a proportional increase in current. The proportionality constant of this relationship is dependent upon many factors, the most important of which is the gate voltage. Thus, operation in this Ohmic region allows the FET to act as a voltage-controlled resistor. However, this approach comes with many limitations. First, the Ohmic region of the FET is very small, and only holds for small voltages (less than 50mV). This would require excessive gainstaging to maintain signal levels throughout an extended signal chain. Second, and more importantly, the current conduction of the FET is not exactly symmetric; meaning that the effective resistance depends heavily on the phase of the drain[2] signal.

A better choice of voltage-controlled component is a light-dependent resistor (also known as photoresistors or photocells). A light-dependent resistor (LDR) is a two-terminal device with a resistance determined by the intensity of light absorbed by the face of the photocell.

[2]Name of the FET's "main terminal"

An LDR may be configured as a voltage-controlled resistor by optically coupling the photocell to a LED. Feeding a control signal to the LED will change its brightness, which in turn changes the resistance of the LDR. For a GL5506 LDR, the full range of resistance spans 0.3-10.0 kΩ. This configuration is favorable to the FET because it does not require any biasing, and it demonstrates symmetric conduction across all 360^{o} of signal phase. Wrapping the coupled LED-LDR pair with tape is sufficient to isolate the LDR from external light sources.

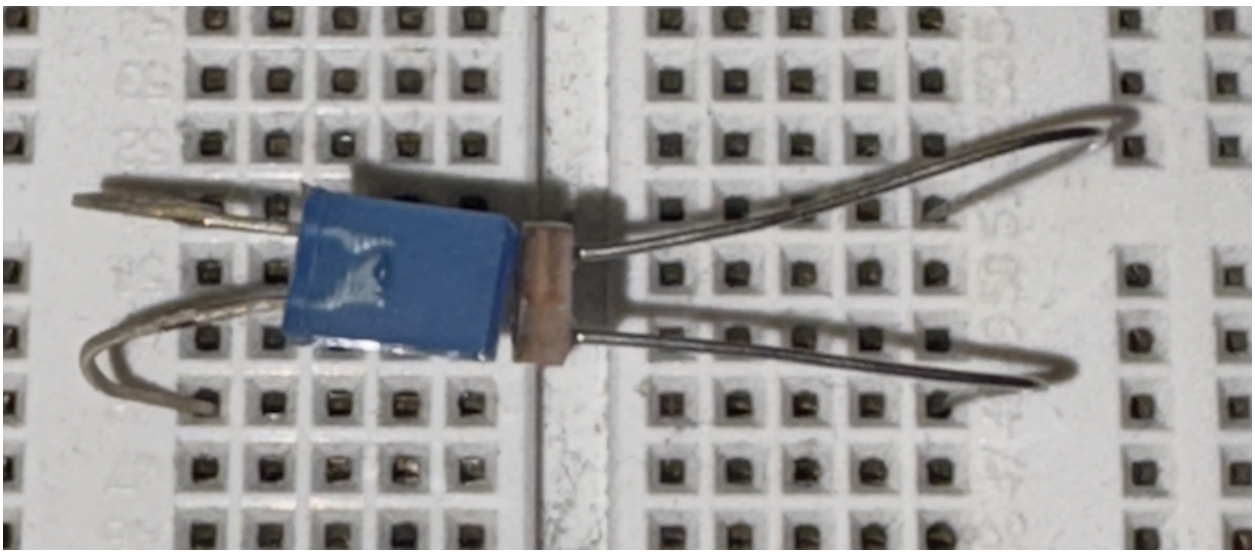

Figure 3: Optically-coupled LED-LDR pair (untaped)

4-Band Equalizer

Foundations

An equalizer is a device which controls the gain of specific frequency bands within an audio signal. Since the timbre of a note is dependent upon the relative amplitudes of the frequencies that compose it, adjustments to the frequency content will result in a change in tone and color. An equalizer works by filtering particular frequency bands and selectively amplifying or attenuating them. These devices may be built from a variety of specific filter types, including high-pass, low-pass, band-pass, and shelf filters.

- High-pass (low-cut): A filter which allows high-frequency components of a signal to pass unaffected while attenuating and applying a *positive* phase shift to low frequencies. The level of attenuation and phase shift depends on how much less than the frequency of a signal component is compared to the filter's cutoff frequency, which is the frequency where the filter applies a -3.01dB gain (the half-power point). The simplest implementation of a high-pass filter is based on a single resistor and capacitor, which determine the cutoff. Steeper cutoff slopes may be achieved by adding additional resistors and capacitors, as well as inductive components (however, inductors are large and bulky, and therefore not very convenient to use in practice).

- Low-pass (high-cut): A filter which allows low-frequency components of a signal to pass unaffected while attenuating and applying a *negative* phase shift to high frequencies. The level of attenuation and phase shift depends on how much greater the frequency of a signal component is compared to the filter's cutoff frequency. Just like the high-pass filter, the simplest implementation of a low-pass filter is based on a single resistor and capacitor, which again determine the cutoff. The key difference between these two filters is whether a resistor or capacitor is in series with or shunts the signal.

- Band-pass: A filter which allows only a specific range of frequencies to pass through it,

while attenuating and applying a phase shift to any frequencies exterior to this range. Such a filter has two cutoff frequencies, marking the upper and lower boundaries of the pass band. The attenuation of frequencies past the filter cutoffs are symmetric, however, the phase shift applied is negative for high frequencies and positive for low frequencies. A band-pass filter may be implemented by cascading a low-pass and high-pass filter (the order of filters does not matter) with overlap between cutoff frequencies.

- Shelf: Shelves are filters which allow low frequencies to pass at a lower level than high frequencies, or vice-versa. The key distinction between a shelf and the filters above is that a shelf will not fully attenuate any frequency. As a result, shelves are characterized by a center frequency (rather than a cutoff frequency) which lies where half of the maximum gain (positive or negative) is applied. Shelves may be implemented by adding series or parallel resistors and capacitors to standard low- and high-pass filters. In audio applications, shelves and high/low pass filters are used primarily on the tail ends of the sonic frequency spectrum, and are often used independently from each other as a result.

- Notch: Notch filters are a specialized form of a band-stop (or band-reject) filter, meaning they attenuate a specific band of frequencies while allowing all others to pass. Generic band-stop filters may be built by placing low- and high- pass filters which have no overlap in cutoff frequencies in parallel. However, notch filters have a highly narrow stop band, which may be achieved with a clever application of phase shifting. This idea is further explored in the discussion of the phaser circuit.

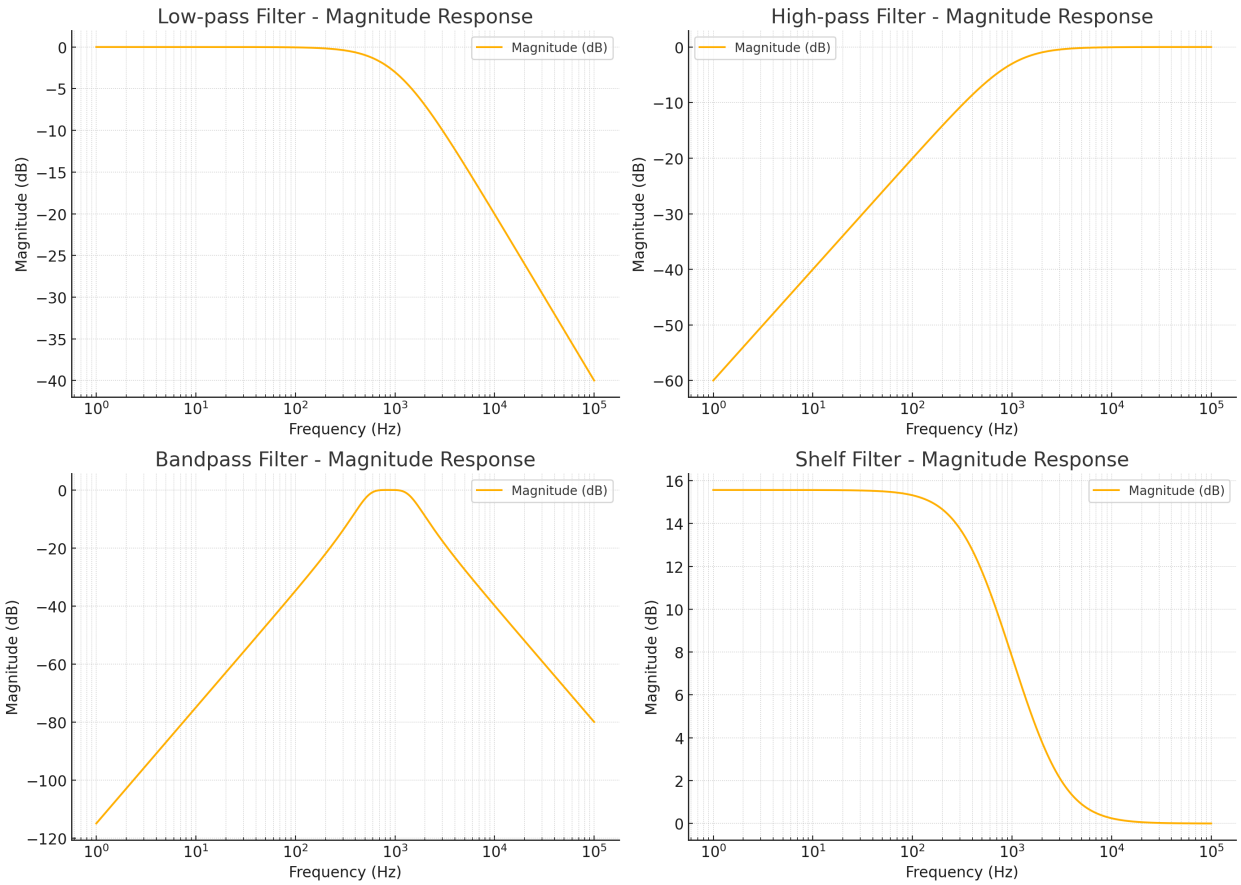

Figure 4: Example filter responses

Filter slope

An immediate challenge in building audio filters is the inadequate cutoff slope of basic filters. Filters constructed with single resistors and capacitors, known as first-order filters, attenuate an incoming signal by about 6 dB for every octave above the cutoff frequency. This is not desirable, since the sonic spectrum is only 10 octaves wide; a high-quality equalizer must offer precise control over multiple frequency bands throughout this range. To illustrate the issue at hand, consider a first-order band-pass filter built from low/high pass filters with cutoffs of 2kHz and 1kHz respectively. This filter will have a frequency response as such:

From the above response, we see that a band with 10dB of dynamic range exists from 315Hz to 6.3kHz. Not only does this band occupy nearly *half* of the sonic spectrum, any amplification or attenuation of this band would affect such a wide range of frequencies that

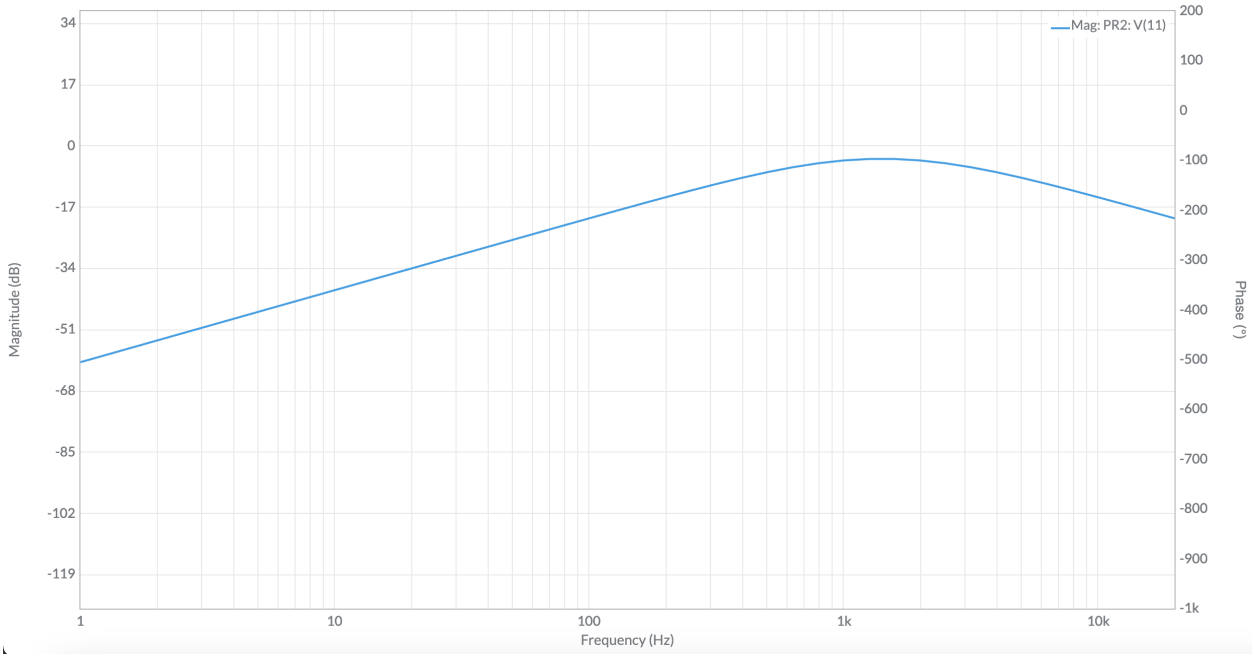

Figure 5: First-order band-pass filter magnitude response

it would only create subtle changes to the timbre of a sound[3]. A first-order filter simply cannot provide sufficient precision to create noticeable changes in musical tone—ideally, we want control over at minimum 1 or 2 octaves per band. Thus, it is necessary to implement higher-order filters.

[3]This is not to say such a filter is useless. For example, applying a boost to about 3-4 octaves of the high end of an acoustic guitar recording may be desirable to brighten its sound without fundamentally altering the familiar tone of the guitar.

Higher-order filters

Higher-order filters are constructed using multiple resistors and capacitors, resulting in steeper cutoff slopes and sharper phase shifts. Since these filters consist of many more components, there exist a variety of different arrangements of circuit components, known as topologies, which each have their own character. In general, every additional level of filter attenuates an additional 6dB/octave past its cutoff frequency.

One of the most common topologies is the Sallen-Key, which is a second-order design characterized by a biquadratic transfer function and a voltage-controlled voltage-source (VCVS). VCVS filters have the advantages of steep cutoff slopes and impedance buffering, meaning that filters can be placed in series while maintaining their tuning. High-order filters also allow for the implementation of feedback loops, which create resonances in the circuit. These resonances appear right before the filter slope, and dramatize the effect of the cutoff. The Sallen-Key topology features a single positive feedback loop from the filters output to its first filter stage, allowing for a small tunable resonance. Ultimately, higher-order filters allow significantly more freedom in their precision, timbre, and implementation.

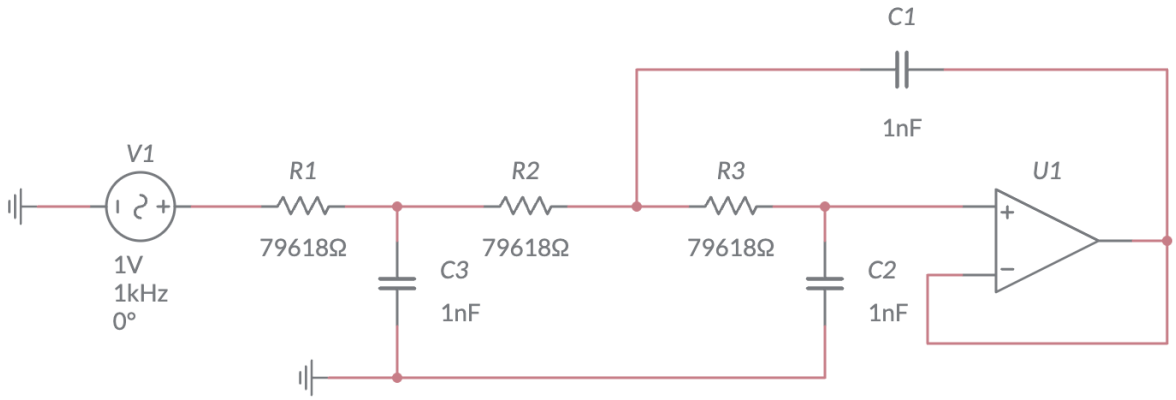

Figure 6: Third-order active filter based on the Sallen-Key topology

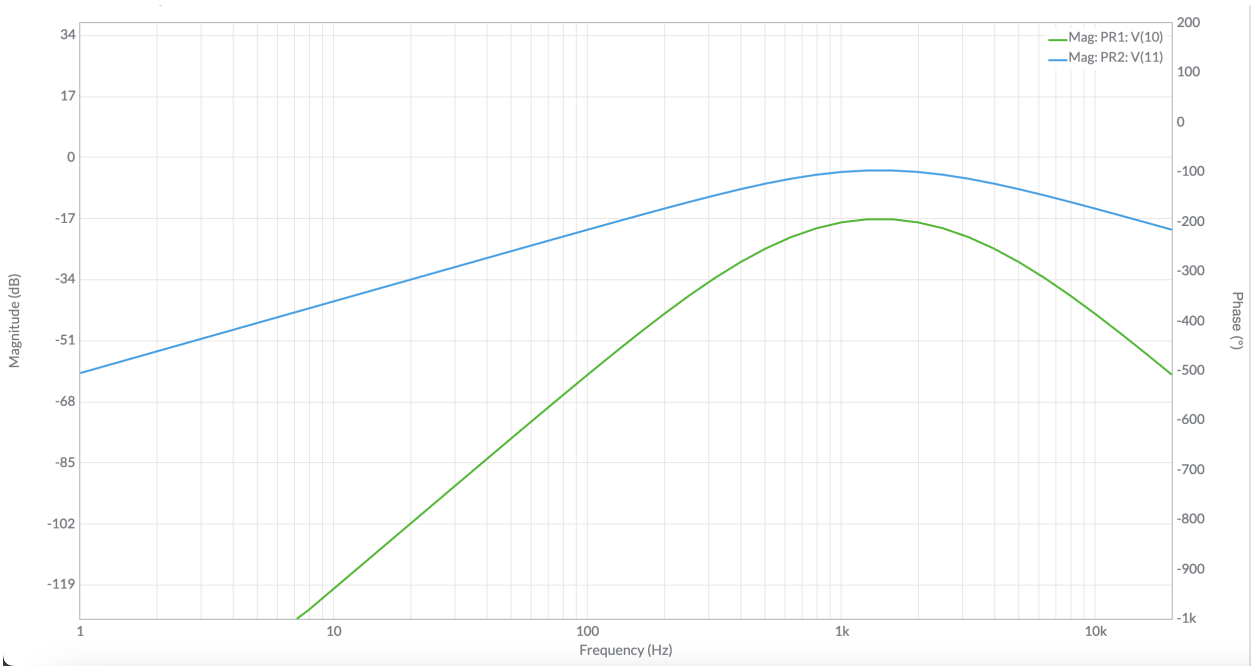

Figure 7: Comparison of 1st and 3rd order bandpass filters

Design

The design of the equalizer uses a high-pass filter, two band-pass filters, and a low-pass filter, each of which are third-order and based on the Sallen-Key topology. The cutoffs of the low- and high-pass filters and pass-band gain of the two band-pass filters are tunable by potentiometers. Shelves were not implemented, as adding shelves would require a much more complex user interface (an additional 4 potentiometer knobs for center frequency and gain).

The equalizer works by splitting the input signal into two bands (250-500 Hz and 2.5-5.5kHz), each of which has an individual gainstage. The lower frequency band was chosen because a boost to this range results in a "underwater" sound—with water being a key element in the aesthetic design of the project. The higher band was chosen for its pleasing presence—boosting or reducing these frequencies can have a strong effect on the personality of a sound. The gainstages are built using inverting amplifiers with a potentiometer in the feedback loop, and can both amplify or attenuate the content of its respective band. Before

reaching the gainstages, a copy of each band is sent to a differential amplifier, which subtracts the contents of each band from the original signal, creating cavities in the frequency spectrum. Next, the gainstage *outputs* are added back into these cavities with a summing amplifier, effectively filling in the gaps. This allows for both the amplification and attenuation of the bands, since simply adding the post-gain bands back to the original signal would allow for only the net amplification of the bands. This summed signal is then sent across a high-pass and low-pass filter in series, each with cutoff frequencies that are adjustable by potentiometers.

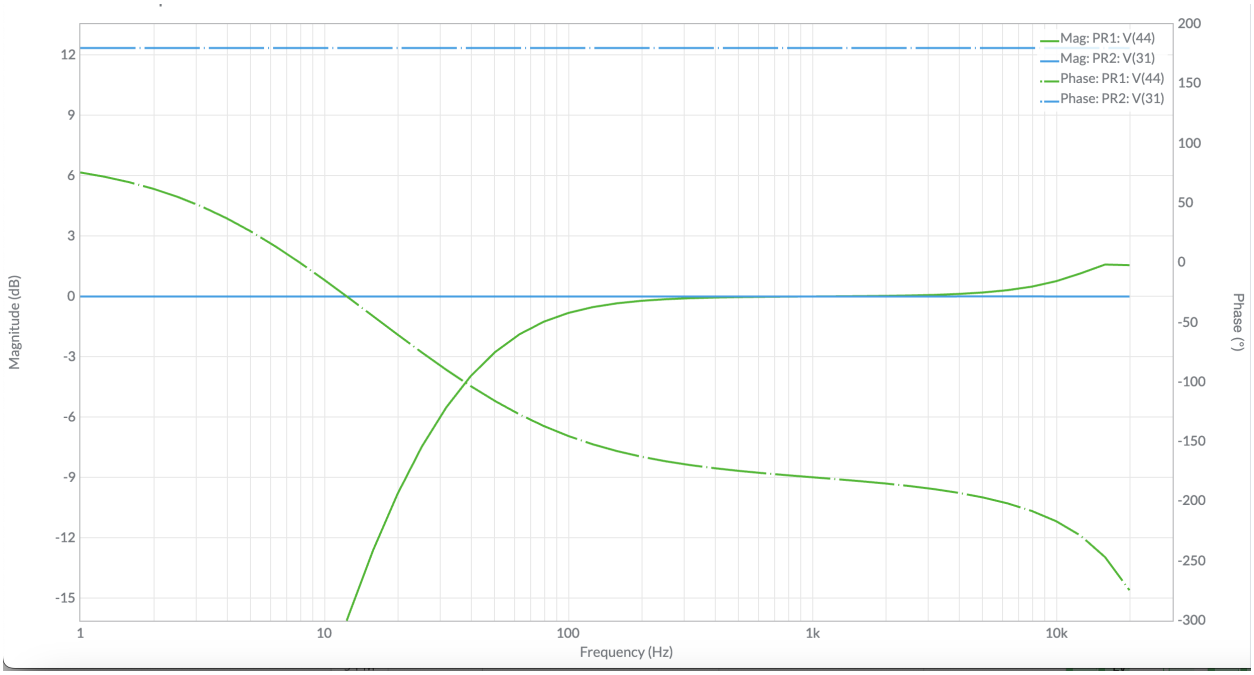

Figure 8: Bode plot, with unity gain for band-pass filters

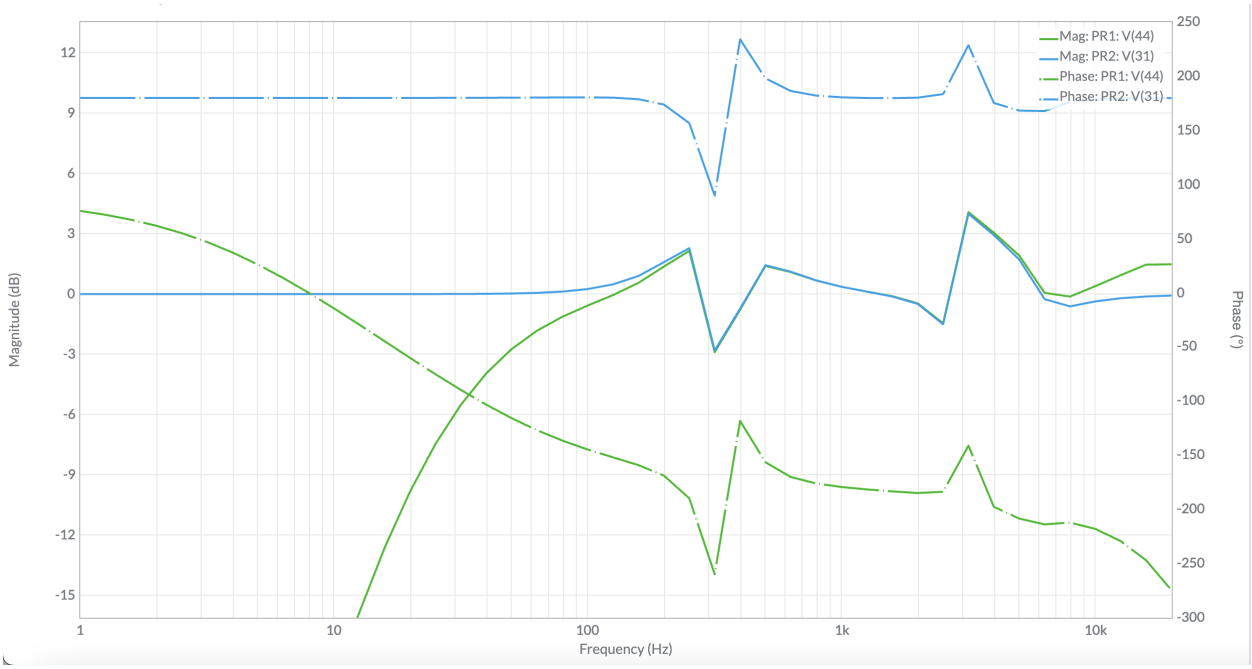

Figure 9: Bode plot, with attenuation for band 1 and amplification for band 2

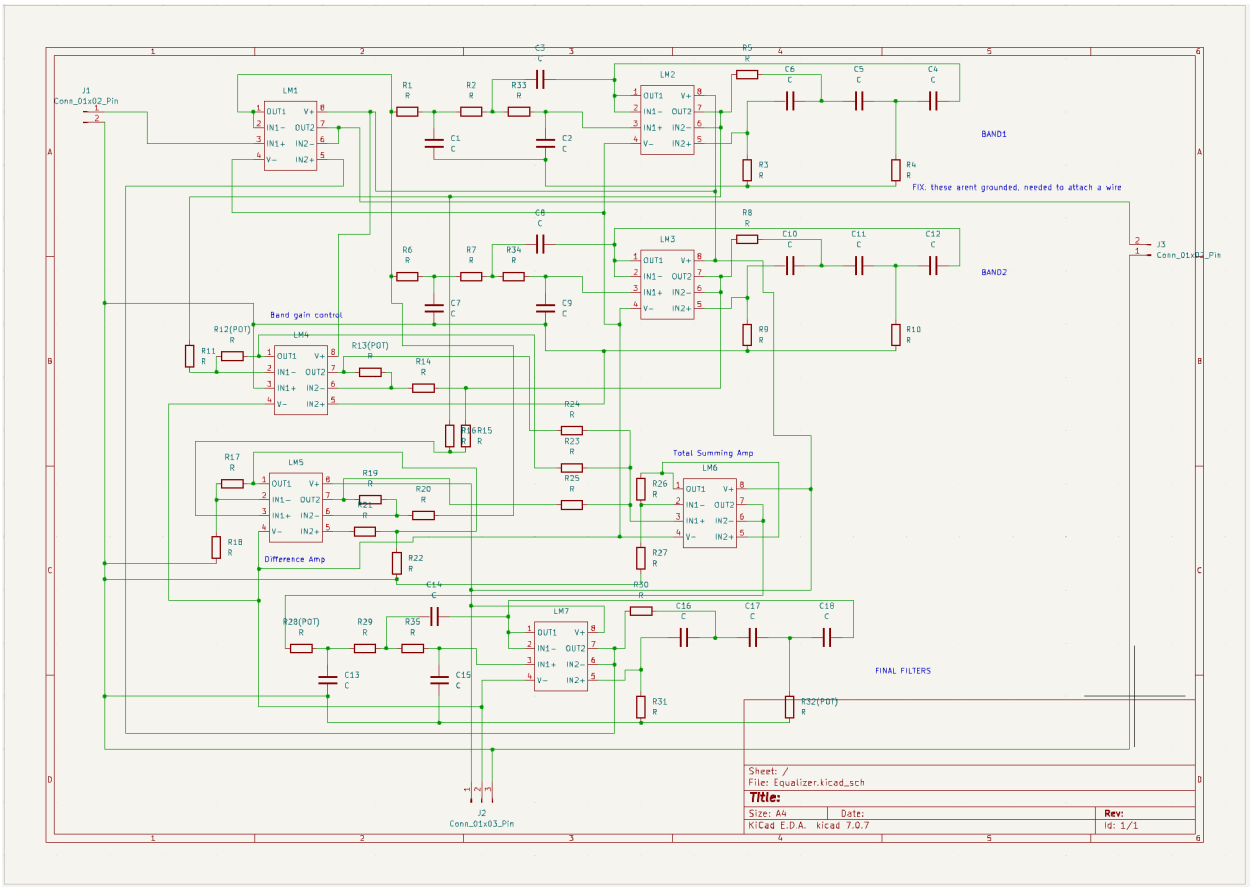

Figure 10: Equalizer schematic

Development and Testing

Resistance and capacitance values for filter cutoffs and feedback strengths were chosen based on simulations run in Multisim. This was done due to the frequency sweep graphing capabilities of the software, which allowed for easy visual comparisons to be made between this equalizer and common commercial equalizers. Ultimately, components were chosen to closely fit standard resistance and capacitance values, as non-standard values would require series/parallel configurations, which would occupy excessive space on the already large PCB.

After achieving a working simulation in Multisim, initial construction and testing was performed on a breadboard using a workbench power supply (+/- 15V). After the circuit was successfully constructed on the breadboard, it was tested using pure tones, white noise figures, and audio signals. White noise is a random and non-periodic signal which has a constant amplitude across the entire sonic frequency spectrum. Since it provides a flat "baseline", any frequency-based effects are easily measured by comparing the "colored" output noise[4] to the baseline signal. Since the equalizer is a spectrum-based effect, testing with white noise and audio signals was most effective, revealing few problems that required resolution before PCB fabrication.

The first issue to stand out was a harsh ringing from the equalizer output, which persisted no matter the gain applied to any band. Because the ringing sounded like a (slightly distorted) pure tone, a fast Fourier transform (FFT) was performed on the output to determine where this tone lay. After probing various network points in the equalizer simulation, it was determined that this pure tone aligned perfectly with a resonance in the high-frequency band-pass filter. The capacitor in the corresponding feedback loop was replaced with a smaller one (larger impedance), and the ringing subsided. This discovery led to the re-evaluation of all feedback loops throughout the circuit, and their corresponding impedances

[4]Interestingly, there exist many different colors of noise that are often used as reference figures for calibration measurements. For example, pink noise is a random signal with power density inversely proportional to frequency. If light had the same power spectrum, it would appear pink. Pink noise is often used to calibrate speakers and public address systems.

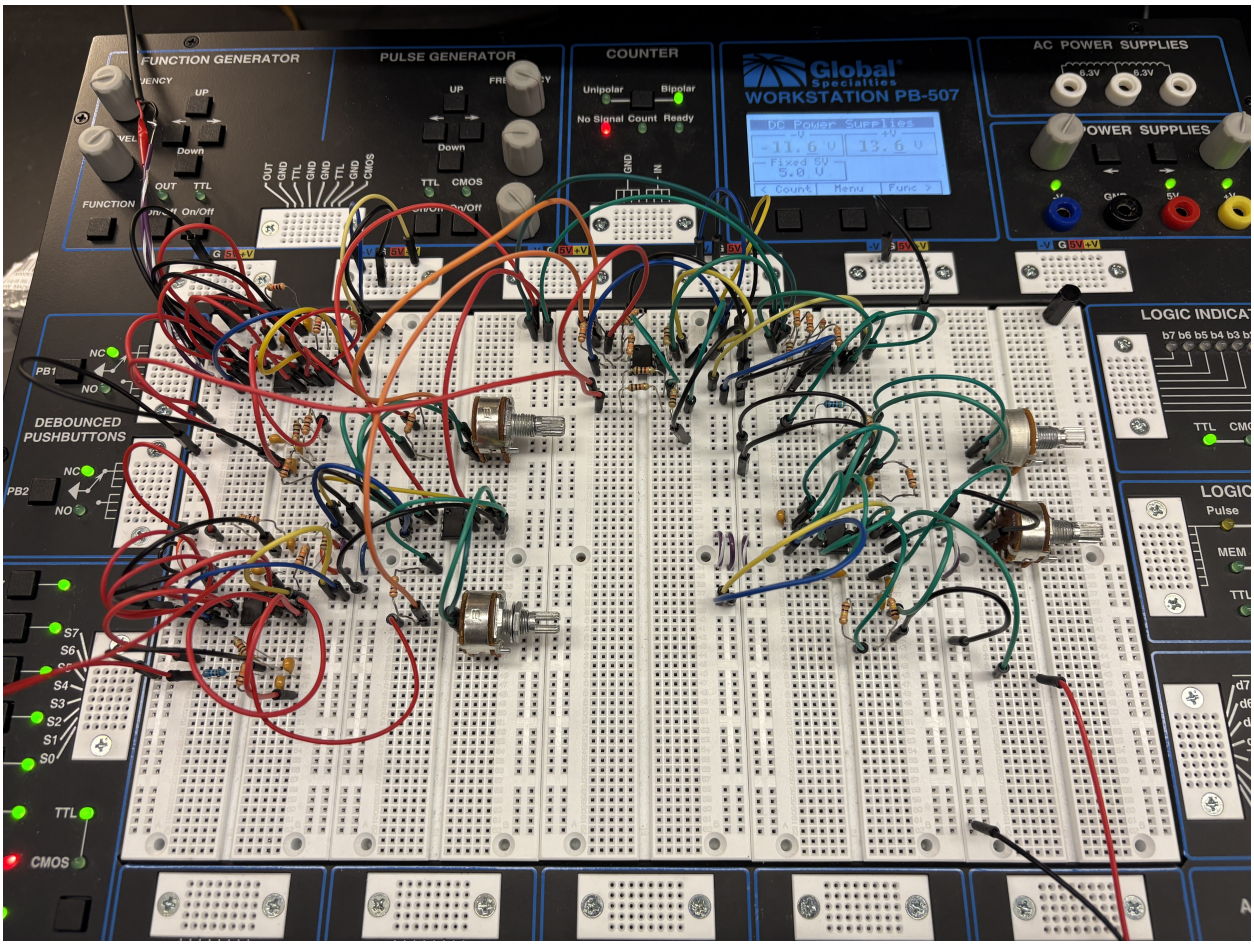

Figure 11: Equalizer prototype assembled on breadboard

were appropriately increased to create sharp, yet pleasant resonances.

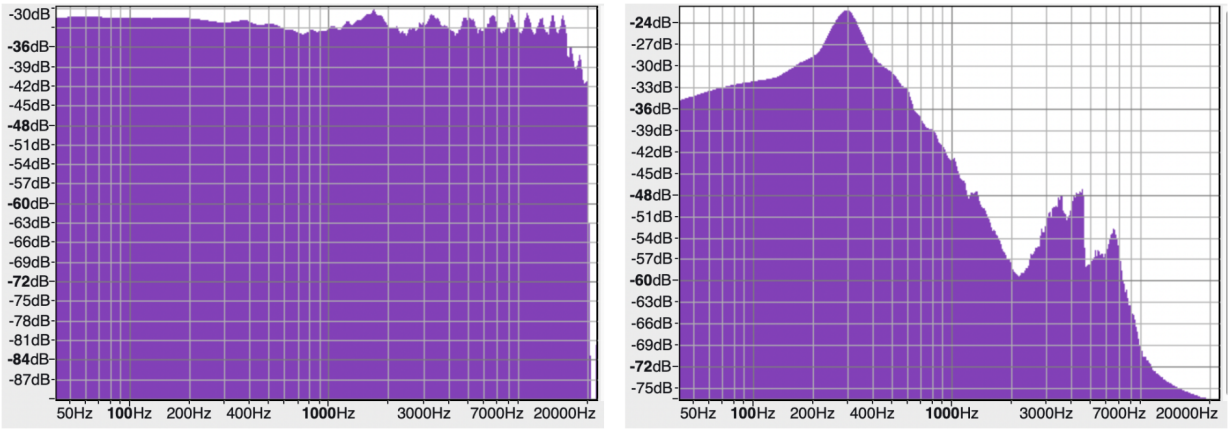

Figure 12: Equalizer output compared to white noise reference

Once the resonance levels were adjusted, the equalizer circuit was drafted in Kicad and a PCB layout was designed. The relevant PCB files were sent to fabrication, and upon the board's arrival, the circuit board was assembled. Initial testing of this board revealed slight inconsistencies with the breadboarded and simulated results, which raised concern. Upon investigation, the Kicad circuit schematic was missing a connection between a filter branch and ground, which was resolved by soldering a new jumper wire connection across the relevant nodes on the rear of the PCB.

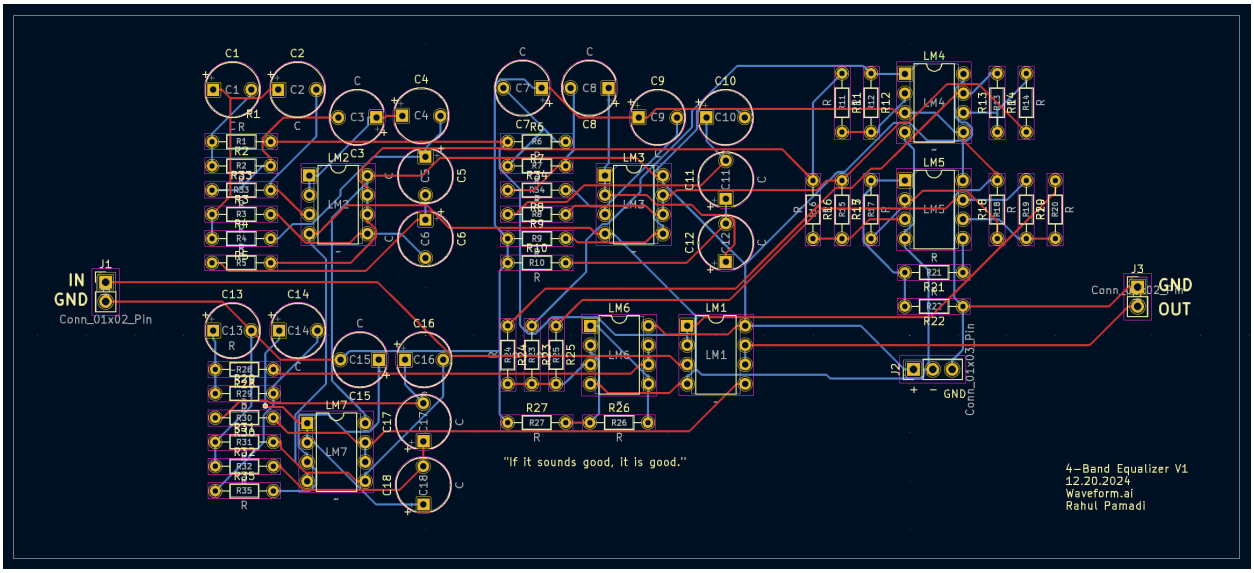

Figure 13: Equalizer PCB

Phaser

Foundations

The aptly-named phaser is a phase-based effect which produces sharp filtering of certain frequencies of a signal. In audio signals, this sounds like a "wooshing" sound effect, and was often used in the psychedelic music of the late 60s and early 70s. This effect is created by summing a signal with a phase-shifted clone; the phase relationship between the two signals results in constructive and destructive interference, which yield notch filtering in the summed signal, and the notch locations may be modulated by an LFO to give the effect a dynamic sound.

Phase shifting

To create this circuit, the first necessary network is one which applies a phase shift to a signal while preserving its amplitude. This is done with an all-pass filter, which can be constructed using a high-pass filter and an op-amp feedback network.

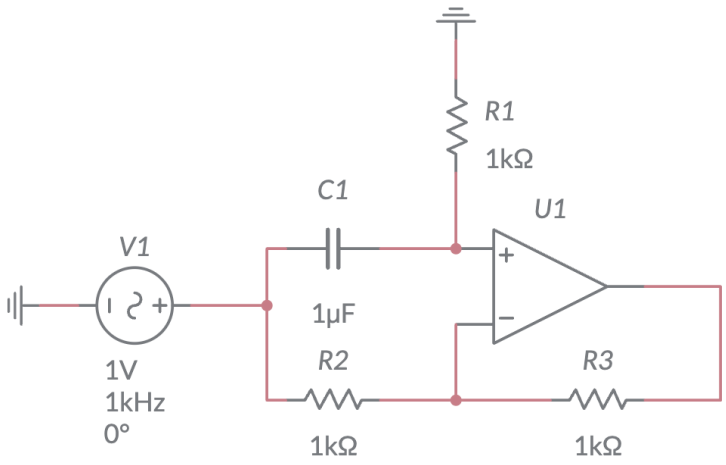

Figure 14: Implementation of an active all-pass filter

Understanding the operation of the all-pass filter is most easy when examining edge cases of frequency input. At frequencies far above the cutoff of the high-pass filter, the capacitor acts as a short, and no current flows the op-amp feedback network. This results in a buffer, and the phase of the input signal is preserved. At the cutoff frequency, the op-amp must apply a 90^{o} phase shift to preserve equal voltages at the inverting and non-inverting terminals, which is fed back to the high-pass filter input, creating partial phase cancellation and thus attenuating the signal to maintain unity amplitude gain. At frequencies far below the cutoff, the capacitor acts as an open circuit, and the network behaves as an inverting amplifier, imparting a 180^{o} phase shift on the input signal.

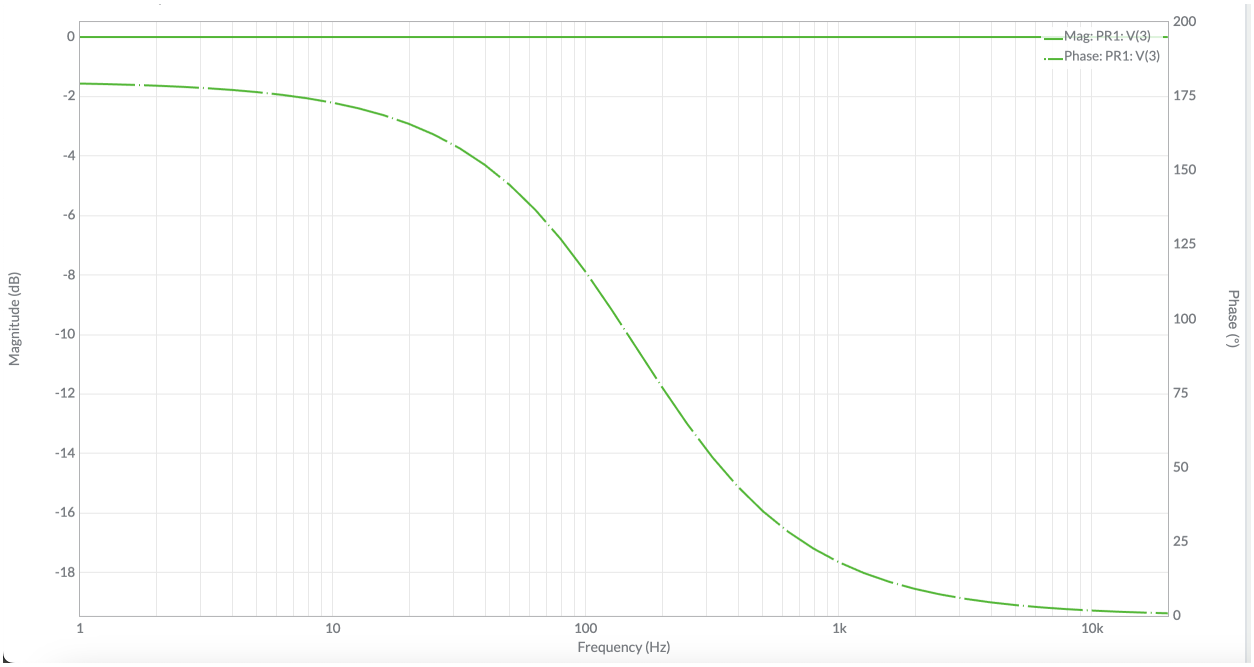

Figure 15: Bode plot for all-pass filter

However, a single all-pass filter is not sufficient to create a notch filter. This is because the range of frequencies shifted by 180^{o} is far too wide; summing this phase-shifted signal with its original will only result in another high-pass filter. In order to create a notch, the maximal phase shift must wrap around to 360^{o}, and a 180^{o} phase shift should only appear at a particular frequency. This can be achieved by cascading two identical all-pass filters in series, which has the effect of simply summing the phase shift contributions of the two filters.

When summed with the original signal, a steep notch is created in the signal's frequency spectrum.

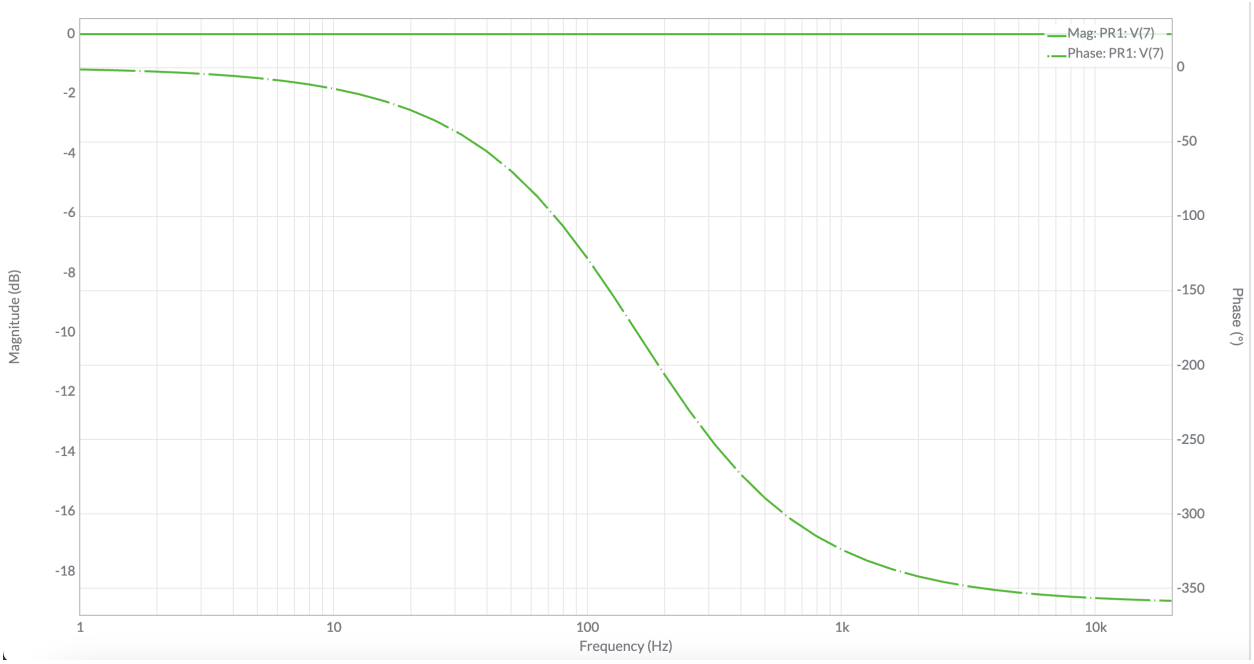

Figure 16: Bode plot for two successive all-pass filters

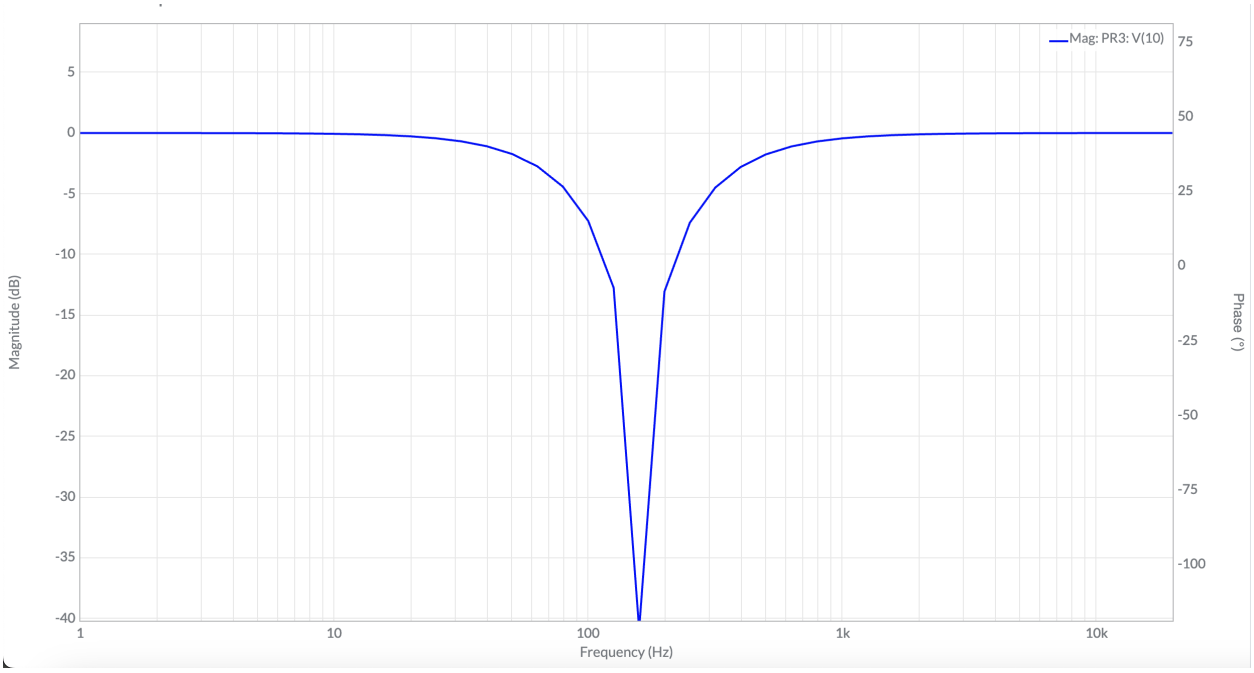

Figure 17: Filtering created by summing the phase-shifted and original signals

Design

The design of the phaser is centered around the notch filter. The center of this notch will lie at the corner frequency[5] of the all-pass filters, which is given by

$$f_c = \frac{1}{2\pi RC}$$

where R and C are the resistance and capacitance of the high-pass filter which the all-pass filter is built from. To modulate this cutoff, the resistors must be replaced by VCRs—either a FET or LDR-LED pair. Since the all-pass filters must have (close to) identical resistance and capacitance values to achieve maximal phase cancellation, care must be taken to ensure that both all-pass stages are modulated in synchronization by a relaxation oscillator LFO. Next, both the phase-shifted and original signals enter a summing stage, where they are recombined to produce the notch filtering. To maintain a unity gain across the entire gain of this summing stage must have a gain of 0.5 to compensate for the addition of the signal onto itself. Since the summing stage must attenuate the two summand signals, it may be built in an active or passive configuration. However, an active summing amplifier is preferred, as it also acts as an output buffer.

Development and Testing

Initial concepts of potential phaser designs were reliant upon the 2N5460 P-Channel JFET as a VCR. Since the characteristics of the 2N5460 were noticed to differ significantly from Multisim, virtual simulations were skipped and drafting was done entirely on the breadboard. However, this FET-based VCR presented many problems—the most notable being a floating offset voltage across the drain and source. This offset persisted even after the addition of 1Mω pull-down resistors configured to shunt the drain voltage to ground. Ad-

[5]The corner frequency refers to the same frequency typically denoted by the term "cutoff frequency". In this context, I use corner rather than cutoff because all-pass filters do not attenuate any signals and therefore do not have a half-power frequency. Corner frequency refers to a more general characteristic frequency of a filter, which in this case is the one shifted in phase by 90^{o}.

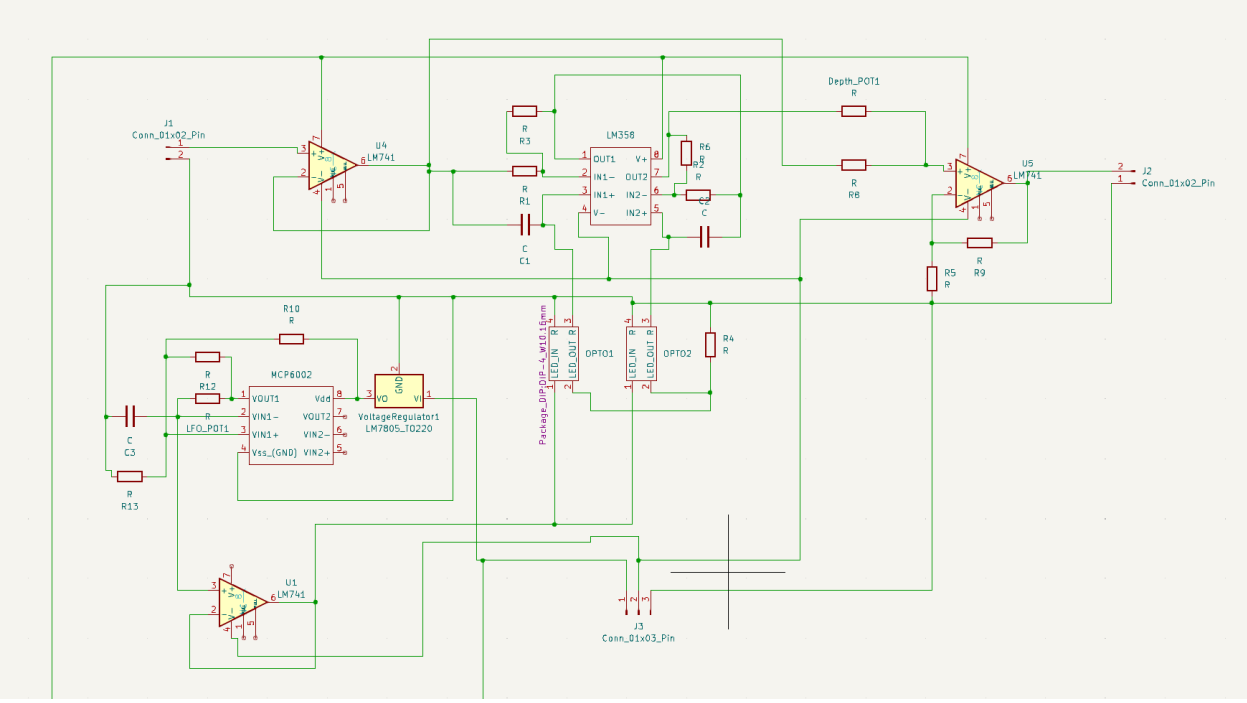

Figure 18: Phaser schematic

ditionally, the transistor showed very little reaction to control voltages at the gate, which implied that it was not operating within the Ohmic region, however voltage measurements at the three terminals aligned with the correct biasing procedures outlined in datasheet figures. Extensive testing was conducted such as swapping 2N5460s, testing with small voltages, and varying configurations of the 2N5460, yet a solution was not found. For these reasons, the FET-based VCRs were replaced with optically-coupled LED-LDR pairs in subsequent designs. This implementation of VCR resolved all of the issues created by FETs and presented none of their own.

Testing was performed on the breadboard, using pure tones, white noise figures, and audio signals. Pure tones were used to test proper function of the all-pass components under modulation. In theory, this will appear on an oscilloscope as a steady wave shifting horizontally according to a control signal. Ideally, the corner frequency should undergo a 180^{o} phase shift at a maximal control signal level, which will only happen if the modulation of the two successive all-pass filters is synchronized. This is exactly what was observed in

testing, indicating the correct behavior of the phasing stage. Like the equalizer, the phaser is a frequency-based effect, so measuring its effect on a baseline white noise figure is extremely useful to evaluate its proper function.

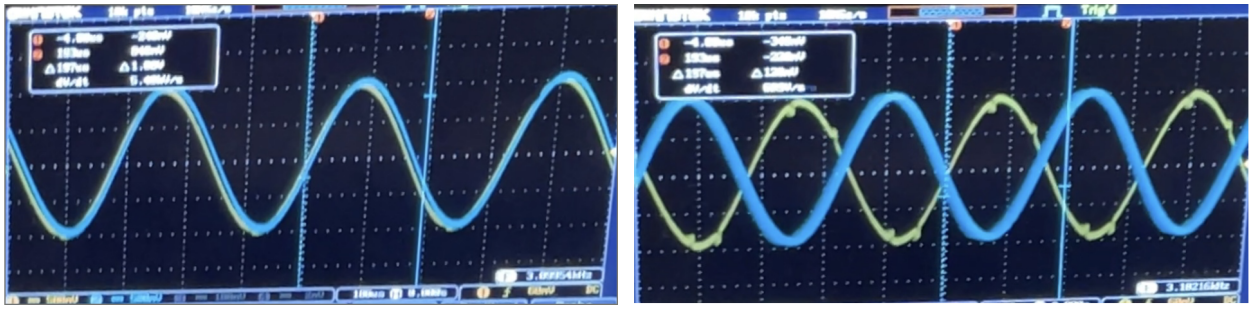

Figure 19: Phase shifting of a near-corner frequency pure tone

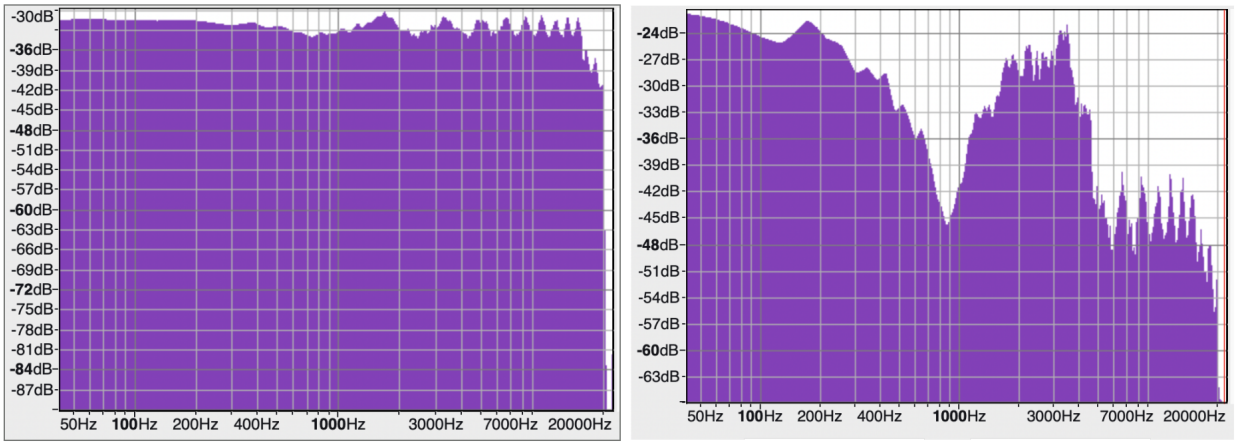

Figure 20: Notch filtering in white noise figure

White noise testing revealed a noticeable attenuation of high frequencies above 7 kHz. Because the attenuation is limited and not sharply rolled off, it is unlikely to result from basic low-pass filtering. Instead, the spectral pattern resembles comb filtering—characterized by a series of notches at harmonic intervals—typically caused by summing a signal with a time-delayed (rather than phase-shifted) version of itself. This suggests a time delay may be affecting the higher frequencies, though the source of this delay remains unclear. Notably, this effect does not appear in simulations using static notch filters, indicating it may stem from modulation effects or physical component characteristics. Regardless of its origin, the

attenuation contributes a pleasing coloration to the sound and was embraced as a analog quirk of the phaser.

A PCB was fabricated in Kicad and assembled. Testing of this PCB yielded very similar results to breadboard testing and needed no further modifications.

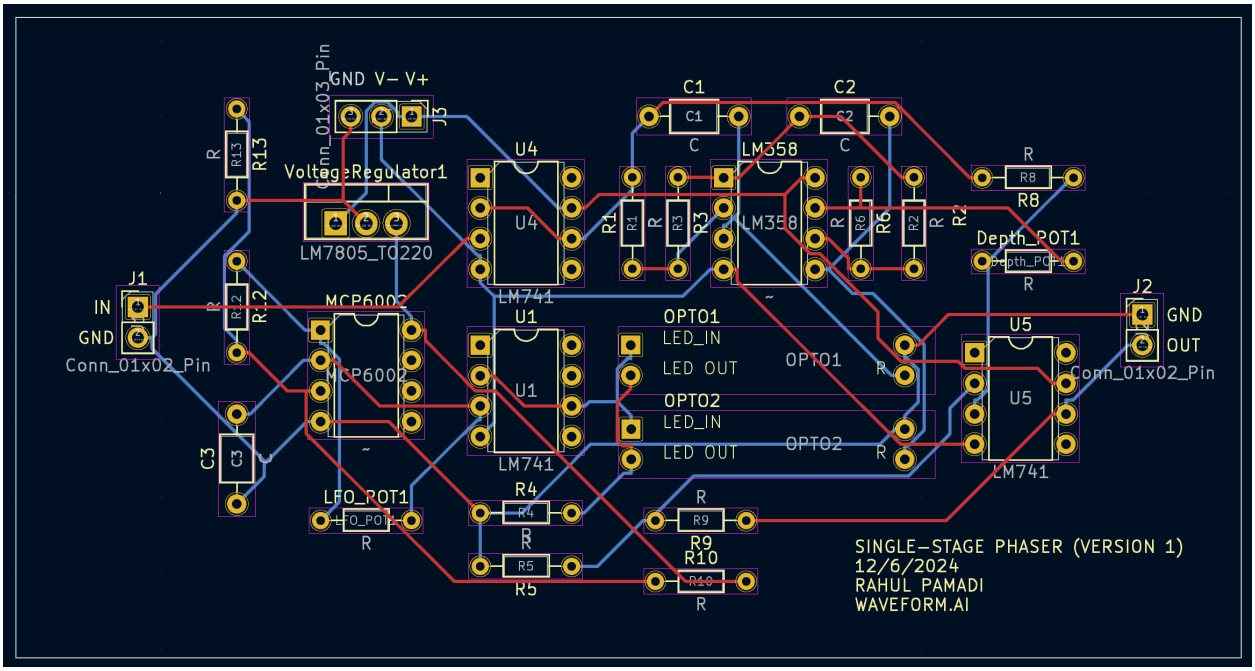

Figure 21: Phaser PCB

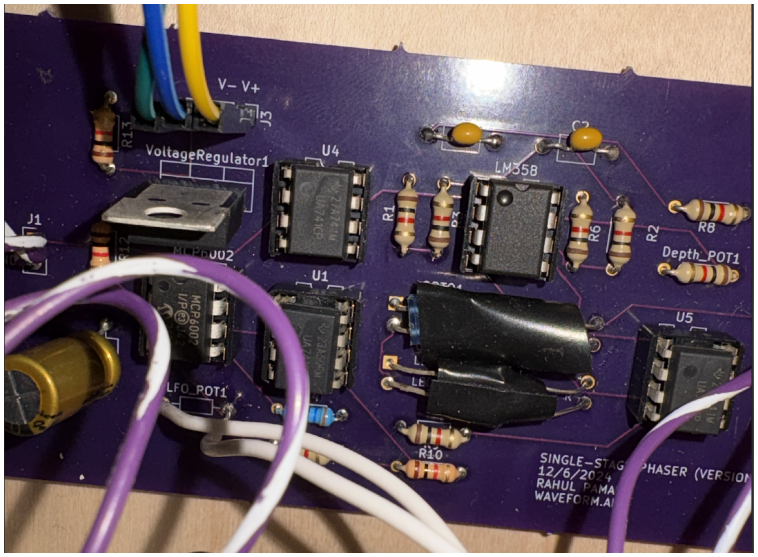

Figure 22: Assembled phaser PCB

Tremolo

Foundations

Tremolo refers to a wavering of the intensity of a sound, commonly created by string players with rapid bowing techniques. Electronically, this effect is achieved by modulating the amplitude of an audio signal over time. The most straightforward way to implement this is by feeding a low-frequency control signal into a voltage-controlled amplifier (VCA), which adjusts the gain of any input signal according to the control signal. This modulation causes the volume to fluctuate, creating the characteristic tremolo sound.

Sidebands

However, while this approach effectively produces the desired wavering effect, it also introduces unintended artifacts—most notably, sideband frequencies. Sidebands are a natural consequence of multiplying two sinusoidal functions (one called the carrier wave, the other called the modulation), which is exactly what amplitude modulation is. More precisely, sidebands arise from the following product-sum trigonometric identity.

$$\cos(\omega_m t) \cdot \cos(\omega_c t) = \frac{1}{2}\cos[(\omega_m + \omega_c)t] + \frac{1}{2}\cos[(\omega_m - \omega_c)t] \tag{1}$$

From this relationship, we see multiplication of two sinusoidal functions is equivalent to the sum of two sinusoidal functions of two new arguments, neither of which is equal to the either of the original arguments. This is undesirable for the purposes of a tremolo, as the original argument (tonal information) is not present in the sum—which creates an audio effect known as ring modulation. For a tremolo, the frequency of the carrier wave must be preserved, and this is achieved by applying a modulation that never assumes a negative value, which preserves the original signal (with the additional sidebands):

$$[1 + \cos(\omega_m t)] \cdot \cos(\omega_c t) = \frac{1}{2}\cos[(\omega_m + \omega_c)t] + \cos(\omega_c t) + \frac{1}{2}\cos[(\omega_m - \omega_c)t] \tag{2}$$

By Fourier's theorem, any (well-behaved) periodic function, such as a polyphony of pitched musical notes, may be decomposed as a series of sinusoidal functions. Therefore the product of this series with a modulation function will result in sidebands for the each frequency across the series. The trigonometric identity above applies only to sinusoidal modulation, and modulation with an oscillation more complex (such as the triangular waves generated by the relaxation LFO) will result in an increased number of additional sidebands. For low LFO frequencies, the tonal effect is negligible, however, as the LFO reaches sonic frequencies, the creation of audible sidebands adds a machine-like whining texture to the audio output. As this quirk aligns thematically with the *Waveform.ai* aesthetic, it was warmly embraced and utilized tastefully in the *Waveform.ai* premiere performance.

Design

Two core circuit networks are required to build a tremolo—a voltage-controlled amplifier (VCA) to modulate signal amplitude, and an LFO to command the VCA. A simple VCA design is as follows: a non-inverting amplifier is built with one of its resistors replaced by a VCR. The VCR is modulated by a control signal from an LFO, which will adjust the gain of the amplifier. A non-inverting amplifier chosen over an inverting amplifier due to the differences in gain created by the ratio of feedback to input resistances.

$$A_{v_{inverting}} = -R_f/R_{input}$$

and

$$A_{v_{non-inverting}} = 1 + R_f/R_{input}$$

Since an non-inverting amplifier may never attenuate a signal and has a baseline gain of unity, the large variation in VCR resistance is softened, creating a less dramatic amplitude modulation.

Development and Testing

Like the phaser, the tremolo makes use of VCRs, and since neither FETs not LDRs are easily simulated in Multisim, initial drafting was performed on a breadboard. The first choice of VCR was the 2N4560 P-channel field effect transistor (FET). Initial testing of this design with pure tone inputs showed promise, with the output signal changing in amplitude in exact accordance with the control voltage. However, upon closer examination, the positive and negative halves of the test output voltage were unbalanced.

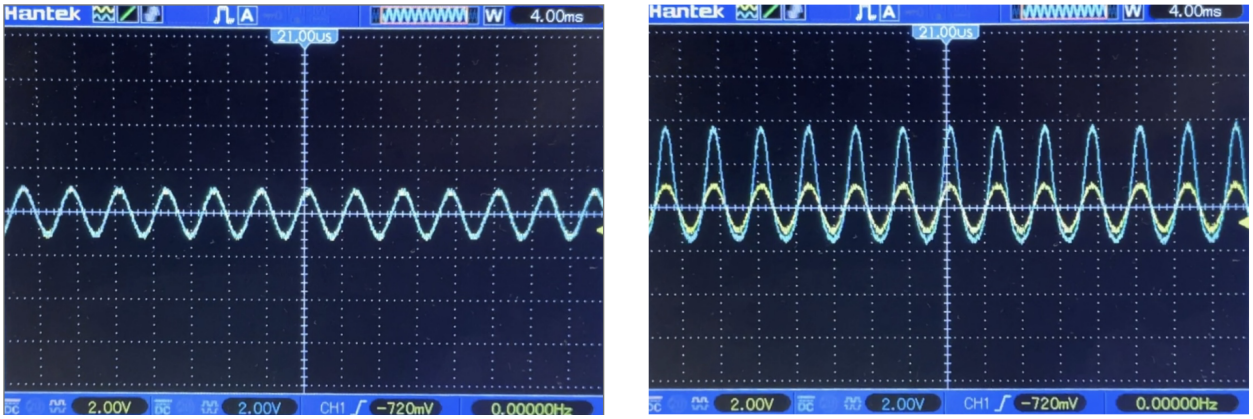

Figure 23: Unbalanced amplification of positive and negative voltage swings

This behavior remained in a range of test frequencies and peak voltages. An FFT of the output signal showed large harmonic distortions (in addition to the sidebands expected from amplitude modulation), which are undesirable in a tremolo effect. This unbalanced behavior was thought to be due to the polar nature of transistors—they are able to conduct more effectively in one direction compared to the other, resulting in greater amplification across the positive swing than the negative swing. This led to a modification of the design using two FETs in the place of the original, each oriented in opposite directions. An additional differential amplifier was added to the control voltage in order to maintain the same gate-source voltage across both orientations of the FET. This solution was ultimately unsuccessful, and while it demonstrated reduced distortion of the output signal, the distortion was still at an unacceptable amount. This is likely due to the individual differences between the

conductivity of the transistors.

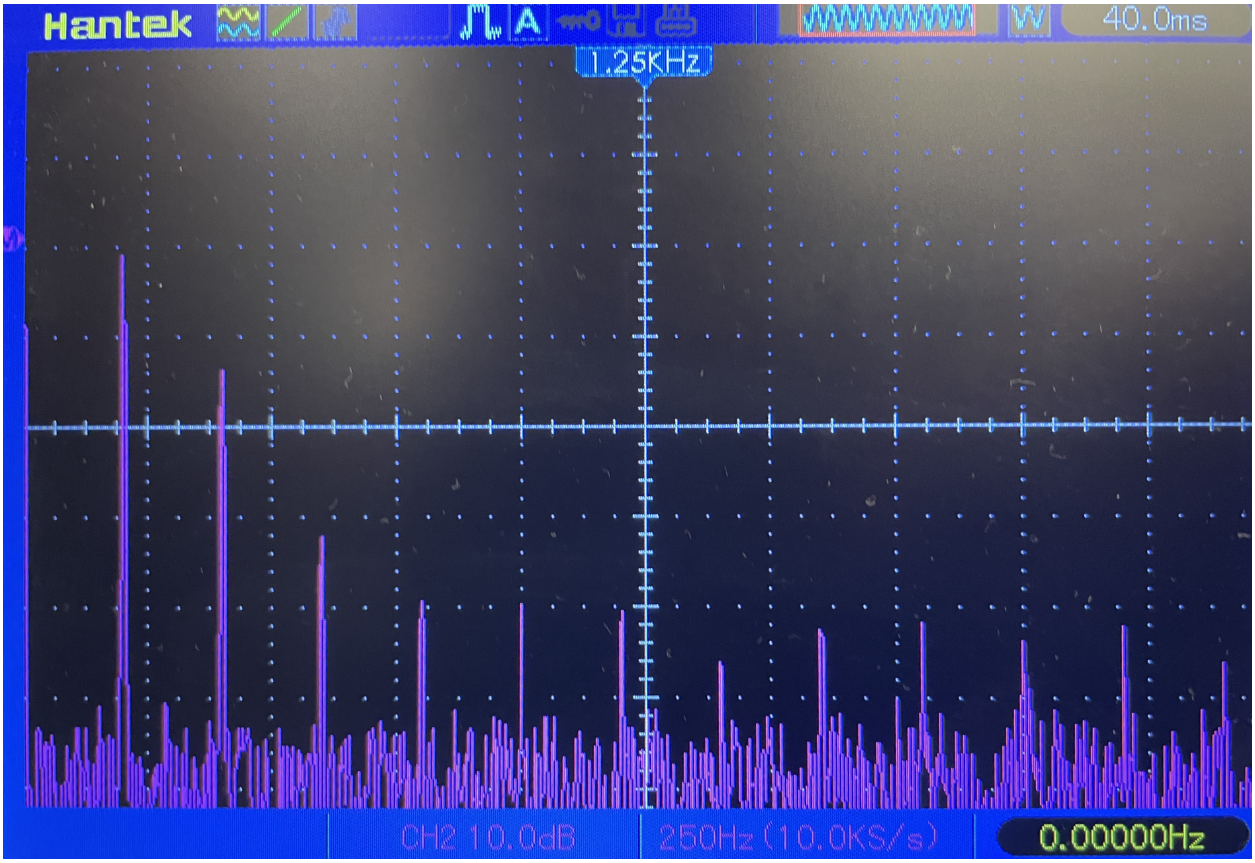

Figure 24: FFT showing harmonic distortion from unbalanced amplification

The next iteration of the tremolo replaced the FET with a unidirectional LDR (GL5506) coupled to an LED, which was driven by the same oscillator (with a slightly higher crossing threshold to compensate for the forward voltage of the LED) as the previous design. In testing with both pure tones and audio signals, the tremolo performed as desired with minimal distortion. A PCB for the circuit was then drafted in Kicad and sent to fabrication. It was assembled, and performed in accordance with all previous testing.

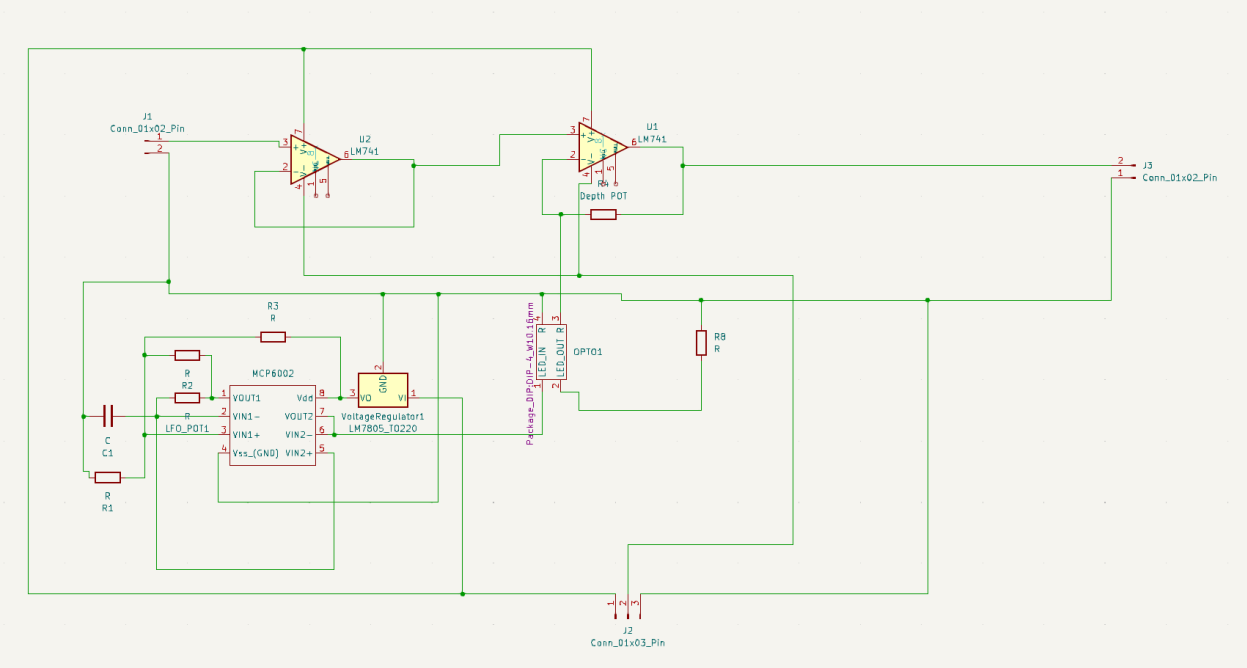

Figure 25: Updated tremolo design, with LDR implementation

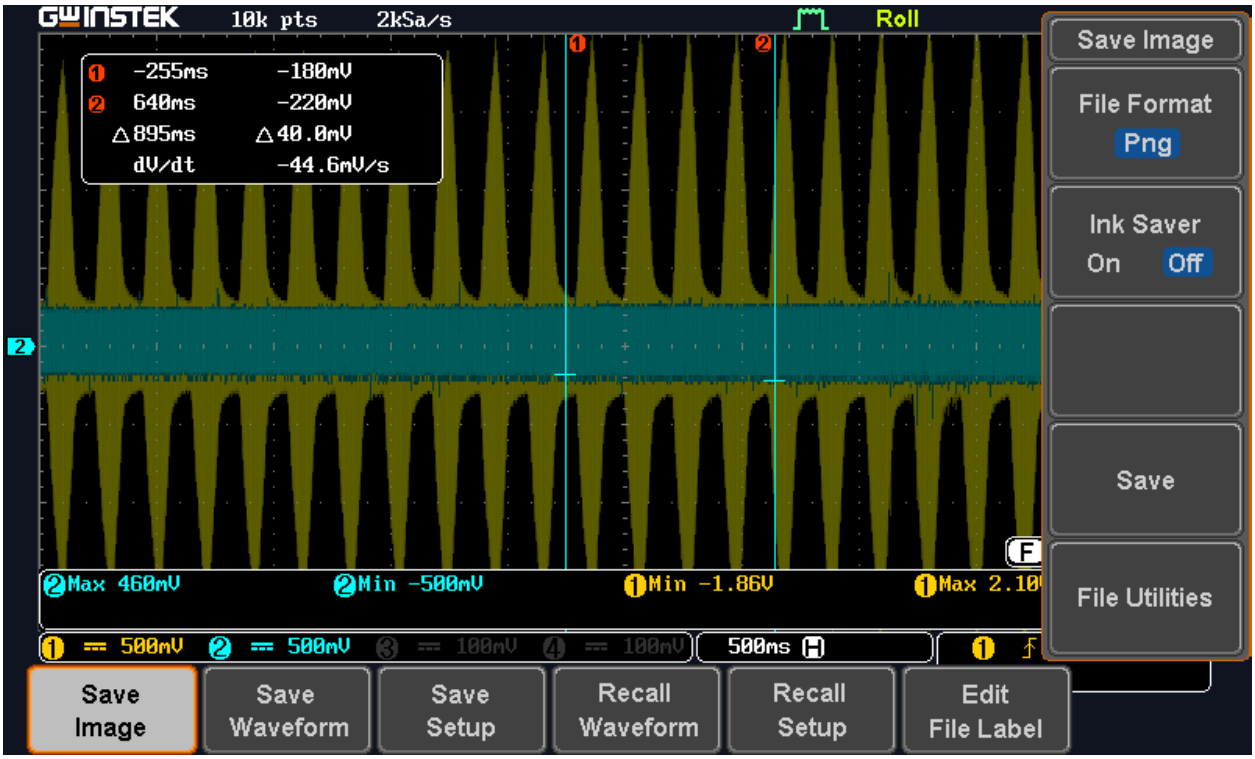

Figure 26: Updated tremolo design input (blue channel) vs output waveform (yellow channel)

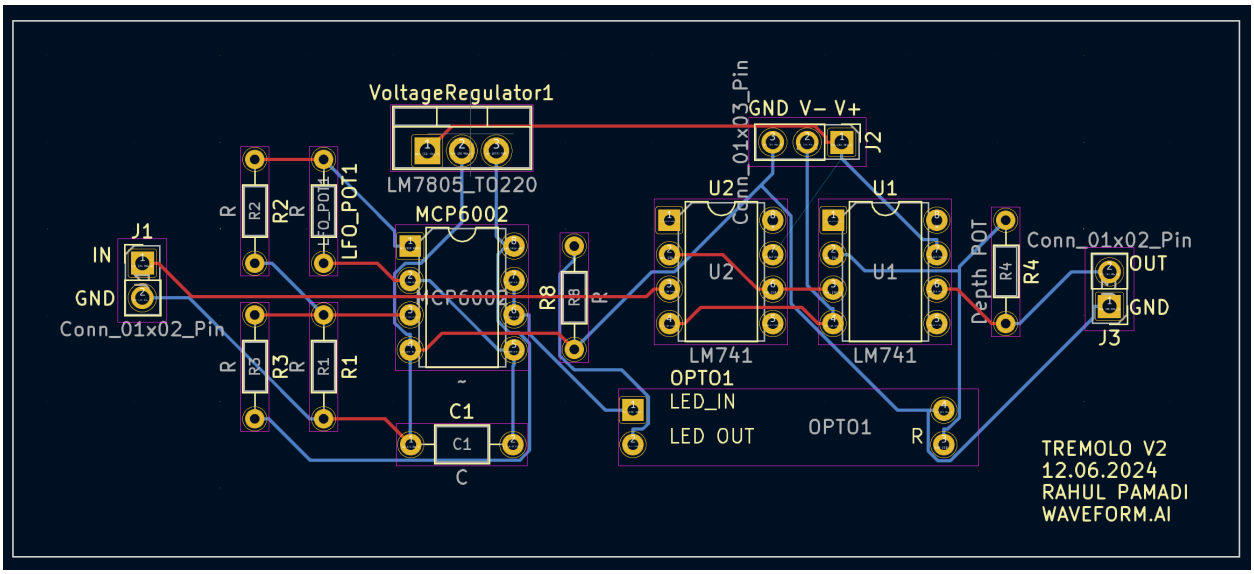

Figure 27: Tremolo PCB

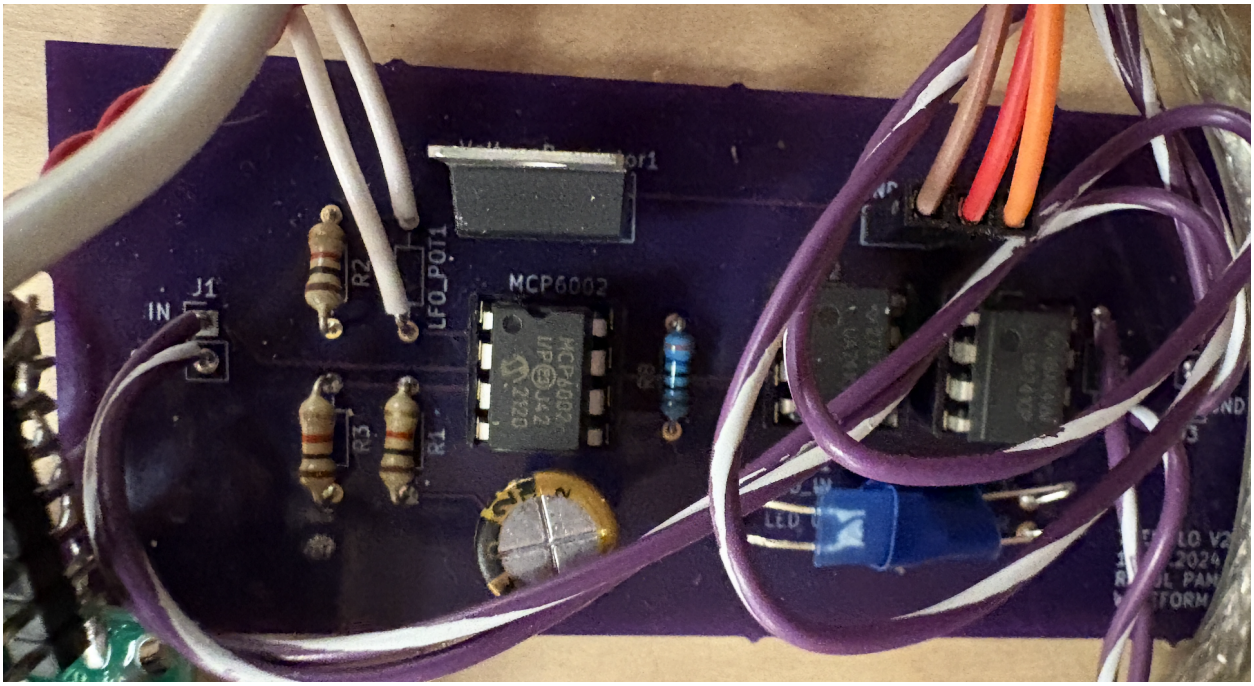

Figure 28: Assembled tremolo PCB

Clipper

Foundations

In audio signal processing, clipping is a form of distortion that occurs when the dynamic range of a waveform is cut off or limited. This often happens unintentionally—for example, in analog systems, it can result from an amplifier being unable to handle a signal that is too strong. In digital systems, clipping occurs when a waveform exceeds a maximum range defined by the bit depth of the system. However, clipping is sometimes desirable. When applied deliberately, the additional harmonic content created by clipping can be musically pleasing, contributing to a sense of warmth, saturation, or richness in the sound.

Hard and soft clipping

A crude design of a clipper can be built from a simple voltage follower. The output of any voltage follower is limited to the supply (rail) voltages of the op-amp it is built with: it may assume any voltage less than or equal to the rail voltage, and any input voltage greater than the rail voltage will simply take the value of the rail voltage. Such a clipper only requires one op-amp to construct, and the clipping threshold is easily adjustable by setting rail voltages with a trimmer. However, the clipping performed by this circuit is very harsh. More precisely, the transition between the unclipped and clipped voltage is nearly instantaneous, leading to cusps in the waveform which introduce excessive harmonic content. This effect is known as “hard clipping”, and is generally perceived as unpleasant. To remedy this, the clipper circuit must allow for a smooth transition between the clipped and unclipped signal. Such clipping is known as ”soft clipping”, and results in fewer harsh harmonics. These reduced harmonics contribute a pleasant warmth to the clipped signal.

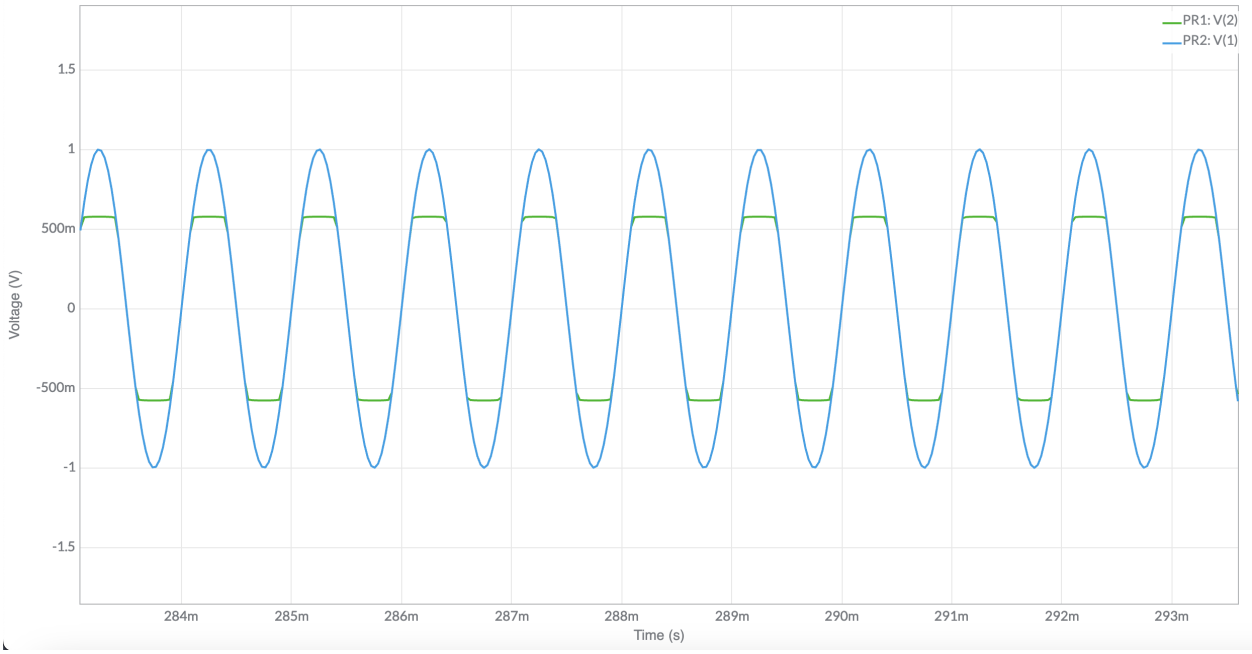

Figure 29: Input signal vs hard-clipped output signal

A soft clipper can be built by exploiting the current-voltage characteristics of a diode. All diodes have a characteristic voltage known as a knee voltage—the voltage at which the conduction of current through the diode is permitted. A diode will prohibit current if the voltage across it is below its knee voltage, and will allow current to flow only when the diode voltage is greater than the knee voltage. This transition to conduction is not instant, and small amounts of current are permitted to pass when slightly below the knee. This soft transition can be used to create soft clipping.

Design

A soft clipper design based on diodes works by using the forward voltage of a diode to short any voltage higher than the diode's knee voltage. However, such a design has an inherent limitation: since the knee voltage is intrinsic to the diodes and cannot be controlled by a voltage, the clipping threshold cannot be changed. In order to circumvent this, a pair of matched amplifiers can be built to amplify an input signal by a given amount, after which the signal is clipped across the diodes. After this stage, the signal is attenuated by the same

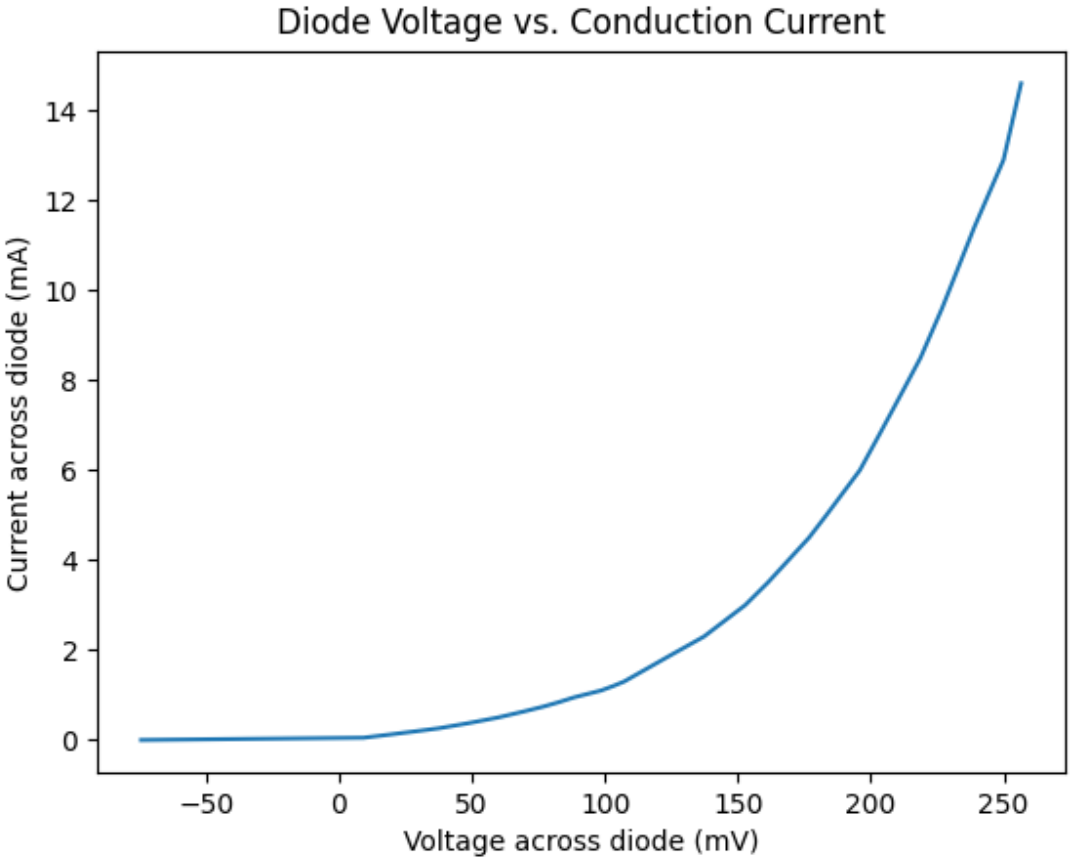

Figure 30: Current-voltage graph in (germanium) diode knee region

amount it was amplified, bringing the signal level to its original amplitude. This matched gain-staging mimics an adjustable clip threshold, giving the user complete control over the degree of clip distortion introduced to the input signal.

The amplification stages are built using two inverting amplifiers—inverting amplifiers were chosen over non-inverting amplifiers due to their ability to attenuate and amplify, rather than just amplify. The total gain of the circuit is given by the product of the gain of each amplifier, and can be written as:

$$A_v = \frac{R_{f1}}{R_1} \cdot \frac{R_{f2}}{R_2}$$

where R_{f1} and R_{f2} are the feedback resistances and R_1 and R_2 are the input resistances

for inverting amplifiers 1 and 2. Since these amplifiers exist solely to adjust clipping level and not alter the input signal level in any other manner, the total gain A_v must be equal to unity. This can be achieved by fixing R_{f1} and R_2 to be the same resistance, which can be done with a double potentiometer. In this configuration, an adjustment to the potentiometer may increase the gain of amplifier 1 and decrease the gain of amplifier 2, or vice versa. Next, for the gain across both amplifiers to be equal to unity and not an arbitrary constant, R_1 and R_{f2} must be set to the same resistance as well. Since adjustments to the clipping threshold are already enabled by the double potentiometer, these two resistances may be left as fixed resistors. With these constraints, the above equation can be written as:

$$A_v = \frac{R_{pot}}{R} \cdot \frac{R}{R_{pot}} = 1$$

Development and Testing

Initial simulation of the clipper circuit was performed in Multisim. Although the software does not support the double potentiometer component, manually adjusting resistors to have equal values achieves the same effect. Once a working model was built and simulated, it was prototyped on the breadboard, allowing for the testing of the double potentiometer. Breadboard testing was performed with pure tones and audio signals,

This circuit was tested with pure tones and audio signals, and performed as expected. The clip threshold was adjustable by potentiometer, and the matched-pair amplifiers maintained a unity gain throughout the unclipped signals. The PCB was then designed in Kicad and sent to fabrication. It was assembled and tested, with all functionality in accordance with prior testing.

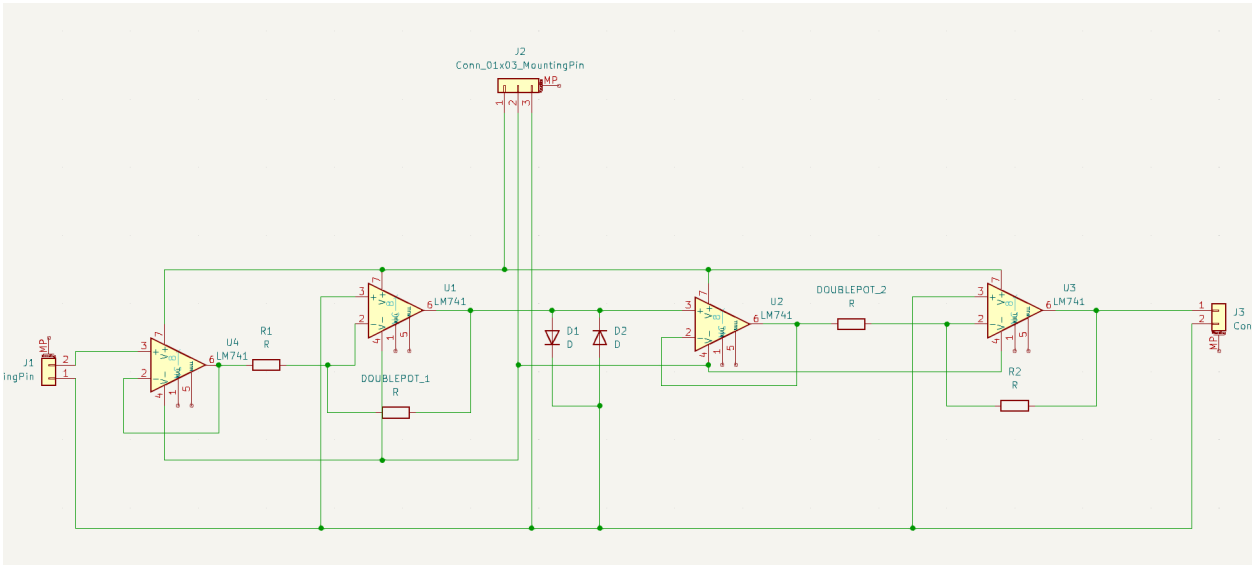

Figure 31: Soft clipper schematic

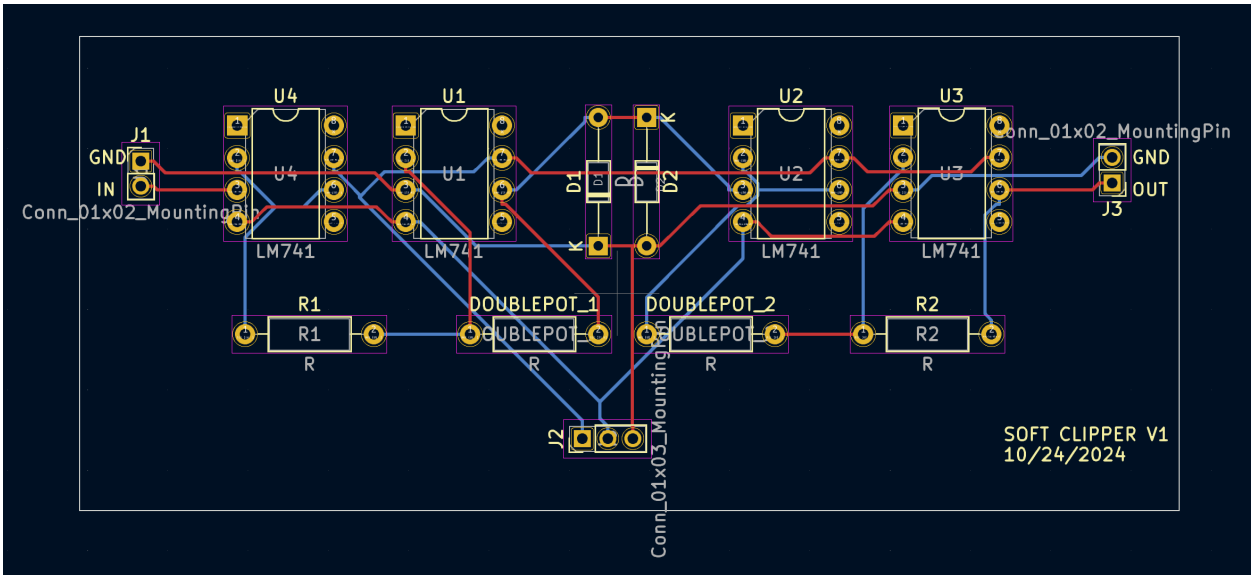

Figure 32: Soft clipper PCB

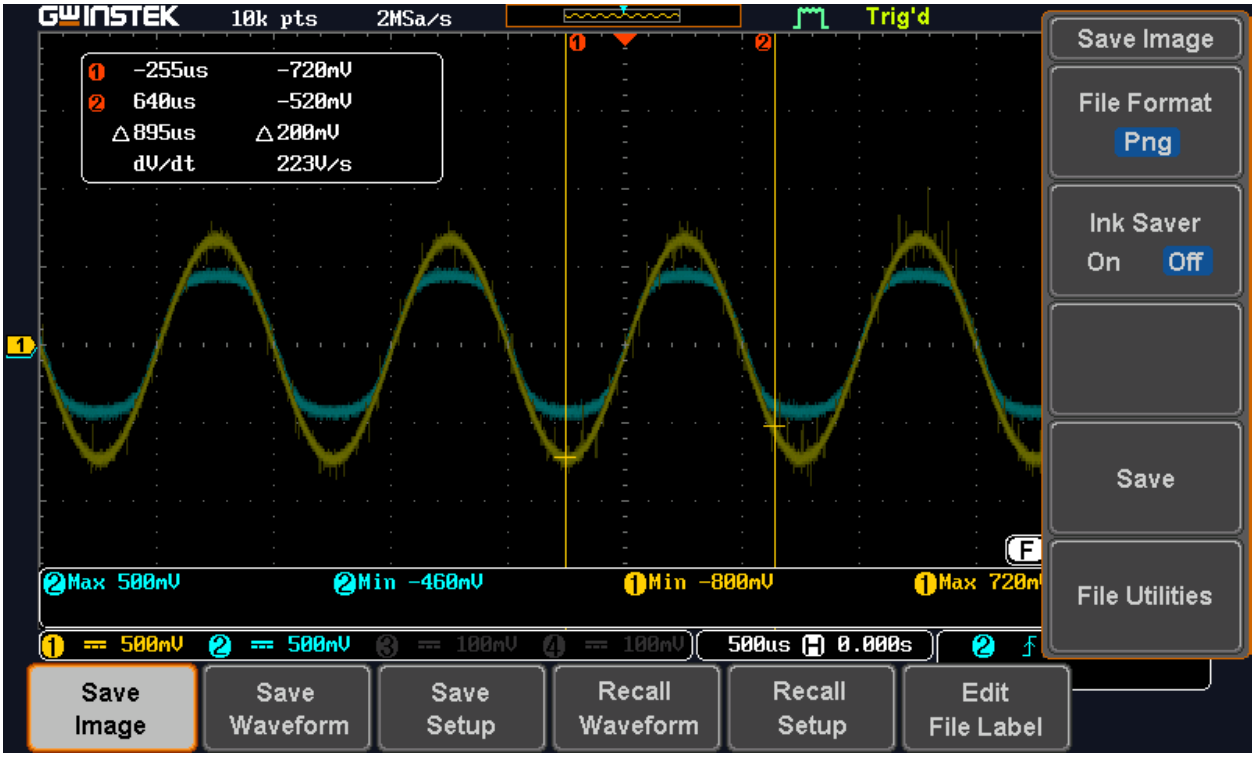

Figure 33: Input waveform (blue channel) vs output waveform (yellow channel)

Wah-wah

Foundations

The "wah-wah" effect, or simply wah for short, is a processing applied to a sound which emulates the characteristic of opening and closing the mouth. This effect is often used by trumpet players by moving a mute in and out of the bell, and electric guitarists playing through a specialized pedal. Acoustically, this sound arises from the filtering (typically band-pass filtering) of certain frequencies by physical barriers, such as the mouth or trumpet mute, which selectively dampen the propagation of spectral content. The timed addition and removal of this spectral content is what produces the wah sound. This effect is achieved by modulating a filter, either mechanically or electronically. For example, in guitar wah pedals, the filter cutoffs are either controlled by a potentiometer tied to the pedal's depression, or automatically by an envelope follower.

Design

Electronic wah effects for electric guitars are most often implemented with a modulated band-pass filter. However, low-end tones generated by a synthesizer are much more important than those of an electric guitar, so excluding them with band-pass filtering would lead to a thin and un-energetic synthesizer output. For this reason, the wah design implements a resonant low-pass filter, which creates the wah effect while preserving the synthesizer's low-end. Additionally, for the sake of the *Waveform.ai* performance, which is aesthetically centered around the undulation of water, the method for modulating the filter was chosen to be an LFO. Such a choice fits the performance theme as well as the previous time-based effects, such as the phaser and tremolo.

The wah consists of two key circuit networks: the filter and LFO. The LFO used is the standard relaxation oscillator, which is optically coupled to a LDR. This LDR determines the cutoff of a third-order low-pass filter, which was designed with a sharp resonance to

hyperbolize the wah effect. As discussed in the previous analysis of filter design, resonances arise from feedback, and the prominence of the feedback is inversely related to impedance across the feedback loop.

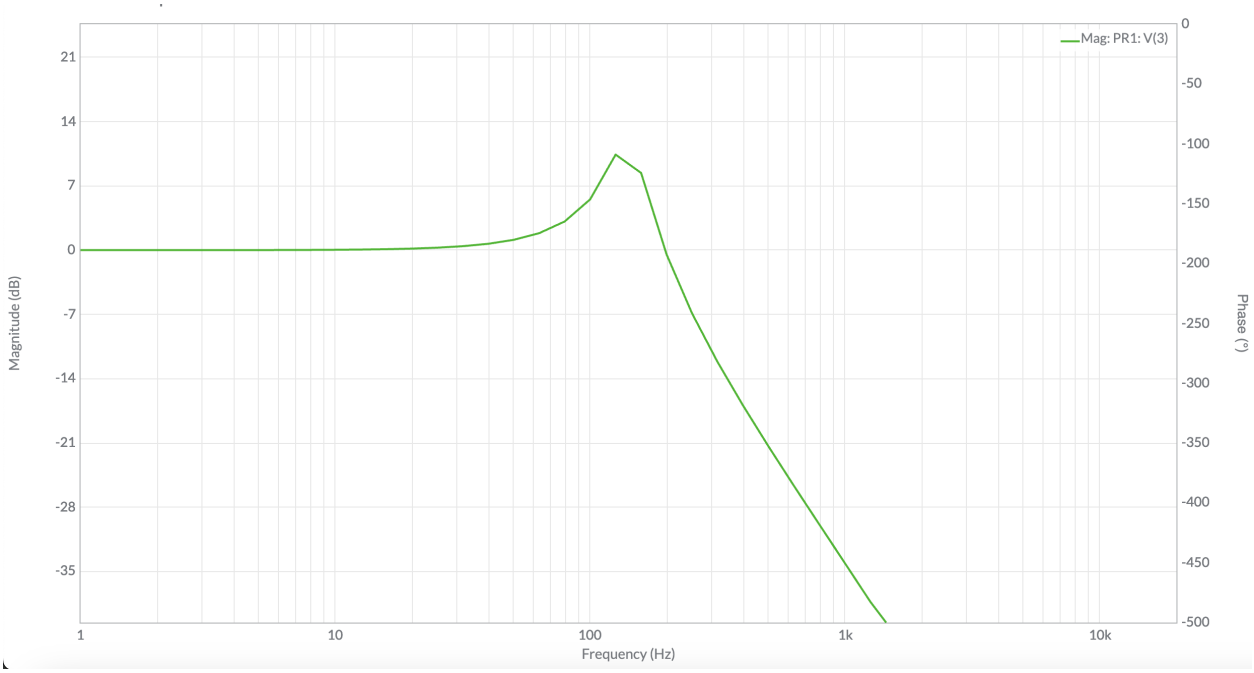

Figure 34: Resonance in 3rd-order low-pass filter magnitude response

Development and Testing

As FET-based VCRs present many technical obstacles, their implementation in the wah was entirely skipped in favor of a optically-coupled LED-LDR pair. Once again, since optical coupling is difficult to simulate in Multisim, all prototyping was performed on a breadboard. Filter component values were determined simply by trial and error—testing with audio signals and determining which cutoff and resonances sounded the best musically. Once a working prototype was assembled and testing on the breadboard, a PCB was designed in Kicad and sent to fabrication. Upon assembly of the fabricated board, it was noticed that a necessary capacitor was missing in the PCB drill. To remedy this, a capacitor was soldered in the appropriate location to the rear of the circuit board.

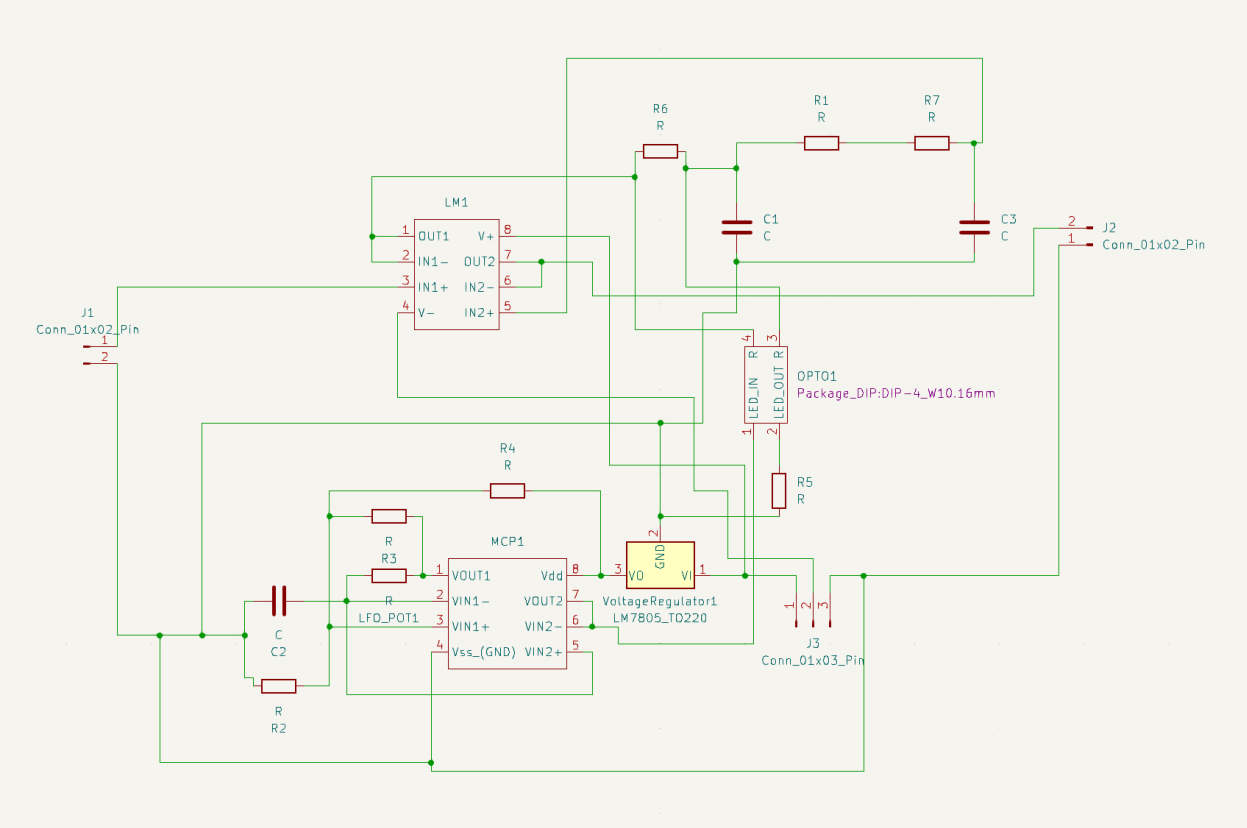

Figure 35: Wah circuit schematic

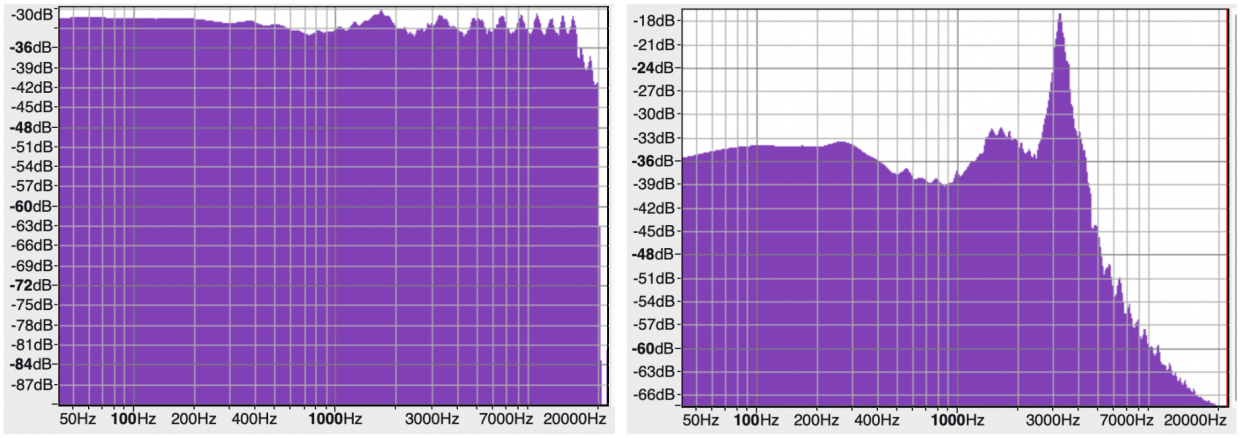

Figure 36: Wah output compared to white noise reference

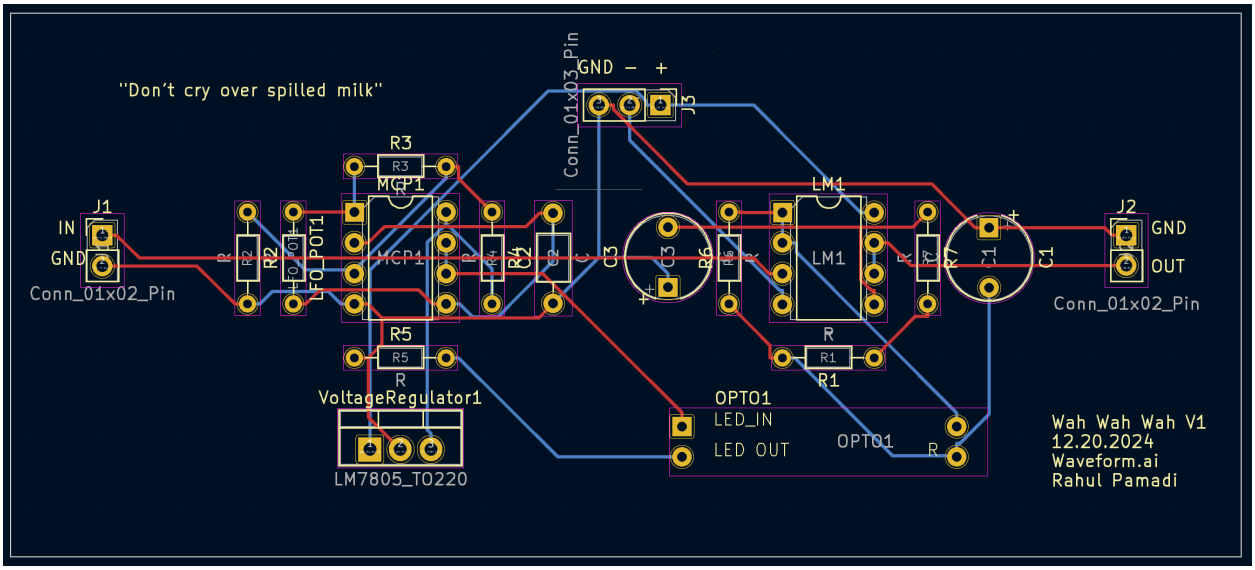

Figure 37: Wah PCB

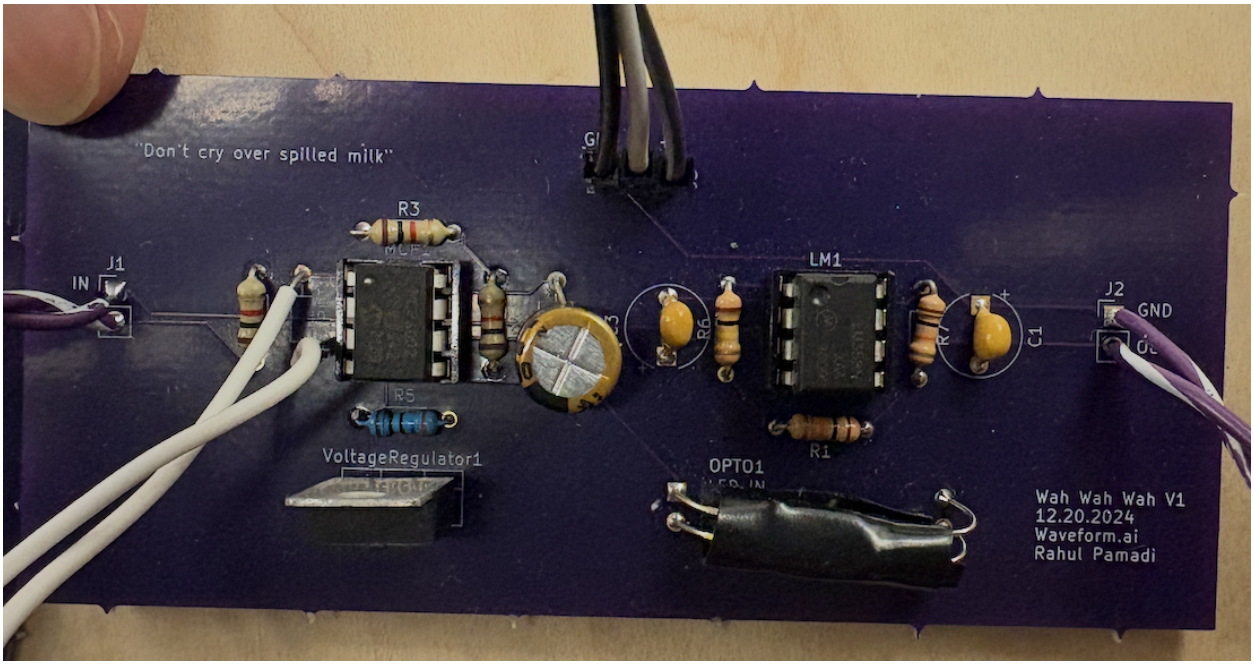

Figure 38: Assembled Wah PCB

System-Level Integration

Although every component effect may operate well in isolation, they must have the capability to integrate within the scope of the synthesizer. This means that the performance of individual circuits must not interfere with the performance of others. The first consideration in system-level testing was gain. Managing gain is important to ensure the complete circuit network does not draw excessive current. Since the tremolo and equalizer are the only circuits which can alter the signal gain, the were tested at extremes to ensure that even at maximum settings, they did not place strain on themselves or the amplifier. Cumulative current measurements were read from a workbench power supply, and did not exceed 0.020 A.

However, excessive current draw was observed when plugging and unplugging patch cables between effects. Additionally, floating voltages at buffer inputs for effects caused buffer outputs to saturate which produced undesired DC flow. This was solved by implementing 6-terminal 1/4 in. jacks, which allow the input pin to be shunted when a cable is not plugged in. This drastically reduced the voltage spikes generated while patching and eliminated floating offsets.

Lastly, when testing the line out to an external system, the noise floor was far too shallow to be sufficient for use with a public address (PA) system. This was solved by implementing a simple adjustable gain stage (built from an inverting amplifier) at the output of the Raspberry Pi.

Failures

Spring reverb

Spring reverb is an analog/mechanical implementation of the specialization effect known as reverb. Reverb emulates the reflections made by sound waves bouncing throughout the physical geometry of a space. A spring box reverb works by transducing an audio signal

through a spring, which "jitters" the wave, mimicking the sound of reflections. Prototype designs of this device resulted in failure, no signal was able to be transduced through the spring and received at the output. This was likely due to the limitations of the piezoelectric transducers from lack of power supplied to them. Building a secondary power amplifier for this device is impractical, since it would require additional communication with the Raspberry Pi to function properly, meaning that it would not easily integrate to the patching framework.

Delay and chorus

Delay is a time-based effect which clones a signal and plays it back at a later time. Chorus is another time-based effect which mimics the sound of a chorus by cloning a signal many times, and adding delays and pitch shifts to the cloned copies. Both of these effects require time-delaying (rather than phase-delaying) a signal, which can only be achieved in analog circuits through a network known as a bucket brigade device (BBD). BBDs are built from a series of capacitors that are charged and discharged in series by transistors according to a clock signal. In order to preserve the fidelity of the original signal, these capacitors must be very small (pF) and the clock frequency must be high (MHz). This means that for any considerable time delay, for instance hundreds of milliseconds, a BBD of thousands of capacitors must be built. For this reason, BBDs are always packaged as integrated circuits. However, BBD chips are no longer mass manufactured, and can only be found from vintage resellers, which introduce issues in procurement (unaffiliated marketplace, extremely long shipping times, etc.).

Conclusion

Although most modern signal processing is performed digitally, analog signal processing still presents remarkable appeal and capability. The subtle imperfections of the medium—either incidental or inherent—may give audio effects a distinct artistic and sonic character. However, while many effects can be effectively realized in the analog domain, others—particularly time-based ones—tend to be more challenging to implement and require precise fabrication or specialized hardware.

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